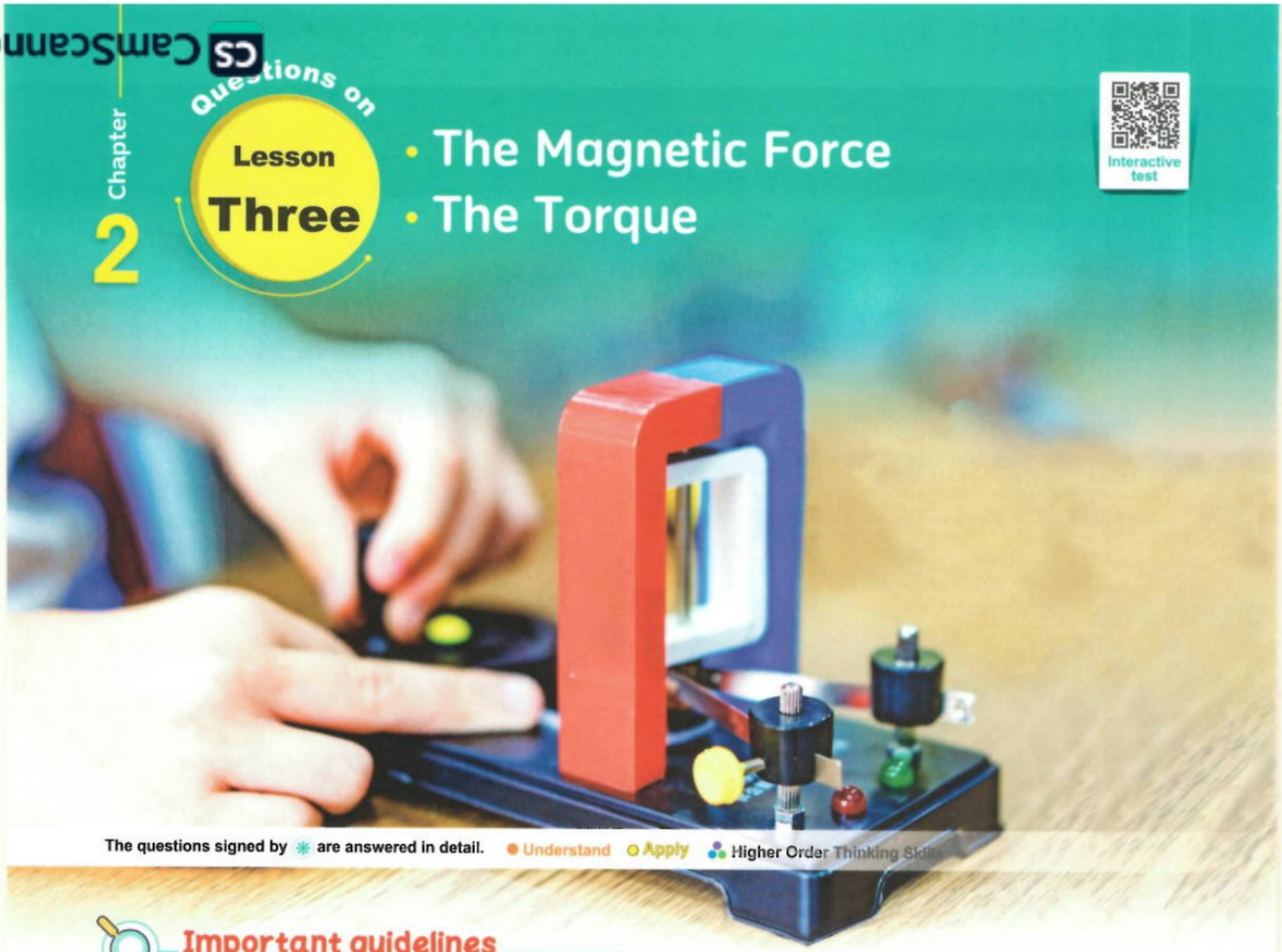
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Chapter 2

Questions on Lesson Three

- The Magnetic Force
- The Torque

Interactive test



The questions signed by ✳ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

Important guidelines

- **The value of the magnetic force (F) due to a uniform magnetic field acting on a current-carrying straight wire:**

$$F = BI\ell \sin\theta$$

(Where: θ is the angle confined between the direction of the field and the current through the wire)

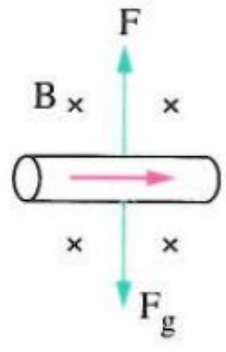
(a) If the wire is parallel to the direction of the magnetic flux:

$F = BI\ell \sin 0 = 0$ (The acting force on the wire vanishes)

(b) If the wire is perpendicular to the direction of the magnetic flux:

$F = BI\ell \sin 90 = BI\ell$ (The acting force on the wire has its maximum value)

- **A current-carrying wire that is placed perpendicular to a magnetic field can be balanced horizontally under the effect of its weight (F_g) downwards and the magnetic force (F) upwards when:**



$\because F = F_g$ $\therefore BI\ell = mg$

$BI\ell = \rho V_{ol} g$ $\therefore BI\ell = \rho A\ell g$ $BI = \rho\pi r^2 g$

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- **The mutual magnetic force between two parallel wires, the distance between them is d and they are carrying two electric currents I_1 and I_2:**

$$F = \mu \frac{I_1 I_2 \ell}{2 \pi d}$$

- If I_1, I_2 are in the same direction, the mutual force is an attraction force.
- If I_1, I_2 are in opposite directions, the mutual force is a repulsion force.

- **The value of the magnetic force by which two parallel wires act on a third wire that is parallel to them in the same plane:**
  - Calculate the magnetic flux density of the first wire at the third wire position:

$$B_{13} = \frac{\mu I_1}{2 \pi d_{13}}$$

  - Calculate the magnetic flux density of the second wire at the third wire position:

$$B_{23} = \frac{\mu I_2}{2 \pi d_{23}}$$

  - Then calculate the resultant magnetic flux: $B_t = B_{13} \pm B_{23}$
  - After that calculate the value of net magnetic force on the third wire: $F = B_t I_3 \ell_3$

Another method:

- Calculate the value of the mutual magnetic force between the first and the third wire:

$$F_{13} = \frac{\mu I_1 I_3 \ell}{2 \pi d_{13}}$$

- Calculate the value of the mutual magnetic force between the second and the third wire:

$$F_{23} = \frac{\mu I_2 I_3 \ell}{2 \pi d_{23}}$$

- Calculate the net magnetic force acting on the third wire:

$$F = F_{13} \pm F_{23}$$

The torque acting on a current-carrying coil if the plane of the coil is:

- **Inclined to the field** → $\tau = BIAN \sin \theta$
- **Perpendicular to the field** → $\tau = 0$
- **Parallel to the field** → $\tau = BIAN$

Where: θ is the angle between the flux lines and the normal to the coil

- **The magnetic dipole moment of a coil:** $|\overrightarrow{m_d}| = IAN = \frac{\tau}{B \sin \theta}$

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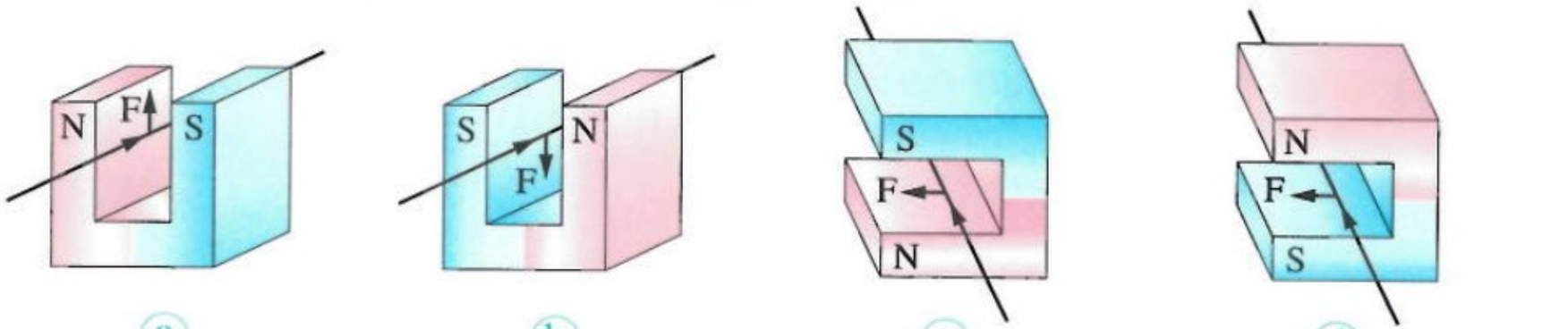
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First Multiple choice questions

Use the following constant if needed: $\mu_{air} = 4\pi \times 10^{-7}$ Wb/A.m

The force affecting a wire

1 In each figure from the following figures, a straight wire carries an electric current and is placed in a magnetic field, which of the following figures correctly represents the direction of the magnetic force affecting the wire?



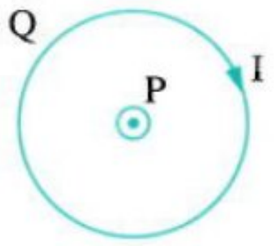
(a) (b) (c) (d)

2 In the opposite figure, a wire of length (ℓ) is carrying an electric current (I) that is directed out of the page. It is placed in a magnetic field of density B whose direction is into the page, then the acting magnetic force on it equals ..... .



(a) $BI\ell$ (b) $\frac{1}{2}BI\ell$ (c) $\sqrt{2}BI\ell$ (d) 0

3 A circular coil (Q) carries an electric current of intensity (I). A straight long wire (P) is placed in the center of the circular coil and perpendicular to its plane as in the opposite figure. If an electric current whose direction is out of the page passes in the wire, then the wire (P) is ..... .



(a) affected by a magnetic force in a direction out of the page
(b) affected by a magnetic force into the page
(c) not affected by any magnetic force
(d) affected by a magnetic force in the plane of the page

4 A wire that carries an electric current of intensity 10 A is placed perpendicular to a magnetic field of density 0.1 T, so the acting magnetic force on a unit of length of the wire equals ..... .

(a) 1 N/m (b) 2.5 N/m (c) 3 N/m (d) 4.6 N/m

5 * A wire of length 10 cm is carrying an electric current of 5 A while being placed in a magnetic field of flux density 1 Wb/m^2, hence the magnetic force affecting the wire, when it makes an angle with the direction of the magnetic flux that equals:

(i) 0°, will be ..... .

(a) 0.3 N (b) 0.11 N (c) 0 (d) 0.7 N

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(ii) 45°, will be ..... .

(a) 0.354 N (b) 0.631 N (c) 0.891 N (d) 0.913 N

(iii) 90°, will be ..... .

(a) 0.25 N (b) 0.5 N (c) 0.7 N (d) 1 N

(iv) 135°, will be ..... .

(a) 0.221 N (b) 0.354 N (c) 0.496 N (d) 0.532 N

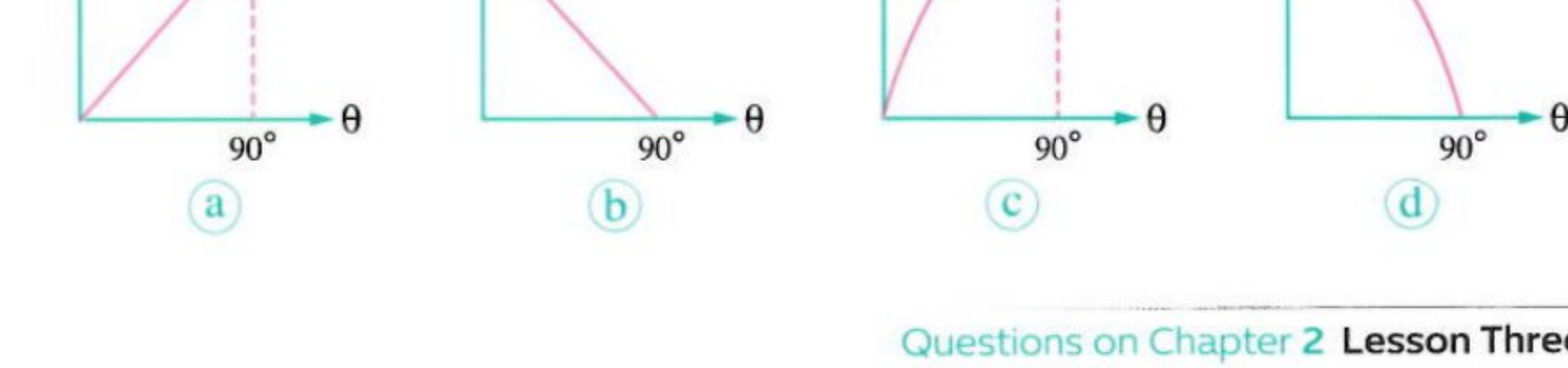
(v) 180°, will be ..... .

(a) 0 (b) 0.25 N (c) 0.5 N (d) 1 N

6 ✳ A straight wire of length 30 cm is carrying an electric current of intensity 4 A, hence the angle at which this wire is positioned in a magnetic flux of density 5 T to be affected by a magnetic force of 3 N equals ..... .

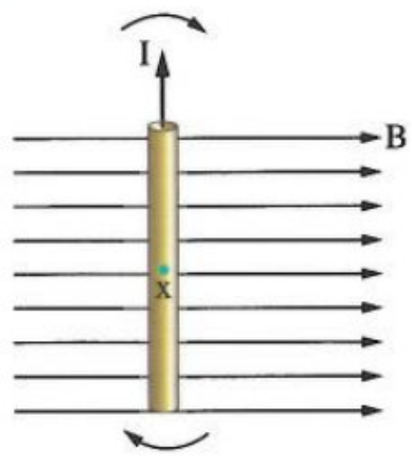
(a) 30° (b) 45° (c) 60° (d) 90°

7 A straight current-carrying wire is placed perpendicular to a uniform magnetic field of a uniformly changeable flux density (B), so the graph which represents the relation between the magnetic force (F) on the wire and the magnetic flux density (B) is ..... .



(a) (b) (c) (d)

8 In the opposite diagram, a straight wire which carries an electric current (I) is placed perpendicular to a uniform magnetic field of flux density B, if the wire is rotated by an angle of 90° about an axis perpendicular to the length of the wire at point x, then the graph that represents the relation between the magnetic force (F) affecting the wire and the angle of rotation (θ) is ..... .



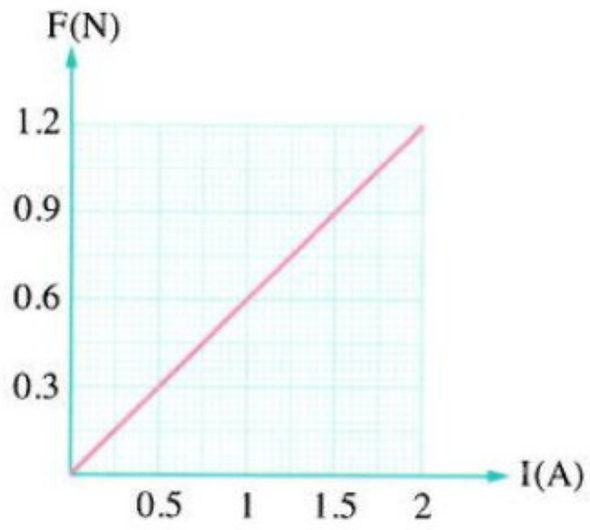
(a) (b) (c) (d)

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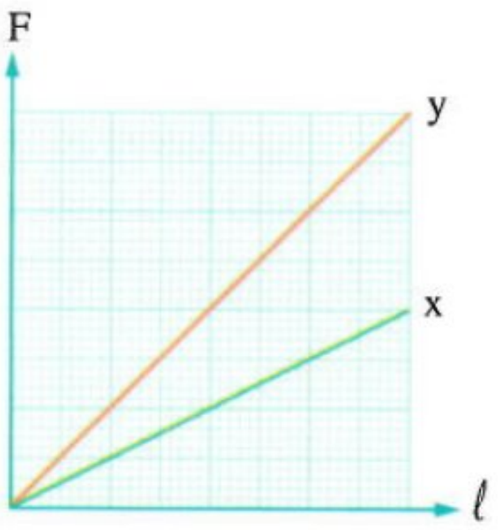
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9 The opposite graph represents the relation between the magnetic force (F) which acts on a straight wire of length 60 cm due to a uniform magnetic field and the current intensity (I) which passes in that wire. The magnetic flux density that acts on the wire equals ..... .



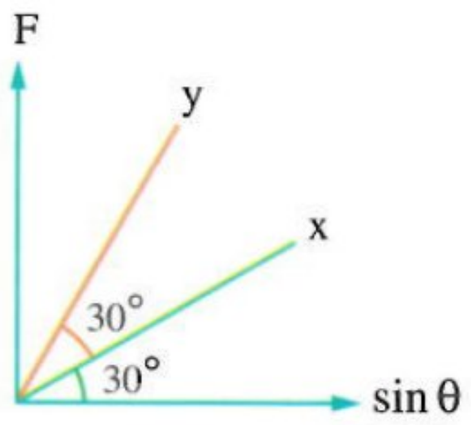
(a) 0.01 T (b) 0.1 T
(c) 1 T (d) 1.01 T

10 Two groups of wires, x and y, carrying electric currents of the same intensity (I) and are placed in a uniform magnetic field of flux density B. The wires in group x are inclined at an acute angle (θ) to the direction of the magnetic field, while the wires in group y are placed perpendicularly to the direction of the field. The opposite graph represents the relation between the magnetic force (F) acting on each wire and its length (ℓ). Then the angle (θ) equals .....



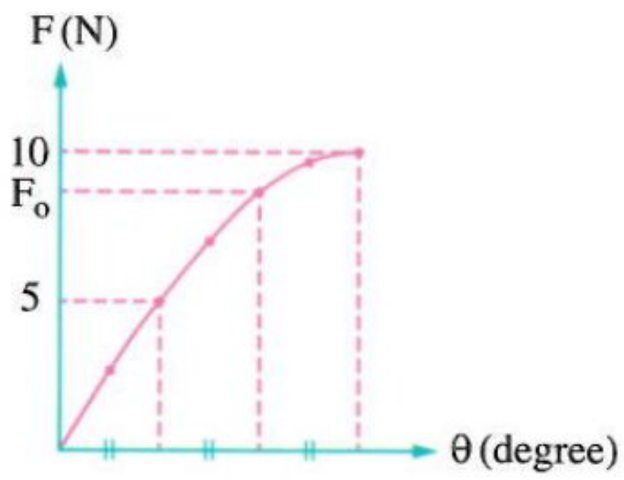
(a) 15° (b) 30° (c) 45° (d) 60°

11 Two wires x and y, of the same length carry electric currents of intensities I_x and I_y respectively. They are rotated in the same uniform magnetic field. The opposite graph represents the relation between the magnitude of the magnetic force (F) acting on each wire of the two wires and the sine of the angle between the wire and the direction of the magnetic field (sin θ). The ratio $(\frac{I_x}{I_y})$ is equal to ..... .



(a) $\frac{1}{\sqrt{3}}$ (b) $\sqrt{3}$ (c) $\frac{1}{3}$ (d) 3

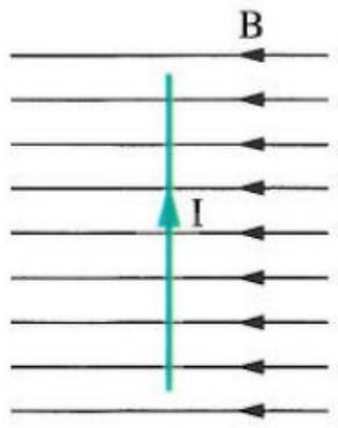
12 A straight wire carrying an electric current is rotated in a uniform magnetic field. The opposite graph represents the relation between the magnetic force (F) acting on the wire and the angle of rotation (θ) of the wire during a quarter cycle from the parallel position to the field. The magnitude of the force (F_o) shown on the graph equals ..... .



(a) $5\sqrt{2}$ N (b) 7.5 N (c) 8 N (d) $5\sqrt{3}$ N

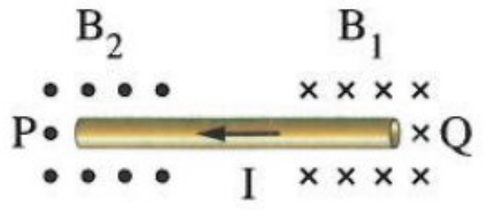
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13 A wire of length 2 m carries an electric current of intensity 5 A perpendicular to a magnetic flux density of 0.4 T and its direction as in the opposite figure, so the value of the magnetic force which is acting on the wire and its direction are ..... .



(a) 2 N, out of the page
(b) 2 N, into the page
(c) 4 N, out of the page
(d) 4 N, into the page

14 The opposite figure represents a straight wire PQ which is free to move. If the wire is in the plane of the page and it carries an electric current of intensity I while two uniform magnetic fields are acting on its terminals, then which of the following choices shows the directions of motion of the terminals of the wire?



	The direction of motion of terminal P	The direction of motion of terminal Q
(a)	Perpendicular out of the page	Perpendicular into the page
(b)	Perpendicular into the page	Perpendicular out of the page
(c)	In the plane of the page upwards	In the plane of the page downwards
(d)	In the plane of the page downwards	In the plane of the page upwards

15 The following figures show a wire that carries an electric current of intensity (I) in different positions in a uniform magnetic field, so the wire which is affected by the maximum magnetic force is in the position ..... .



(1) (2) (3)

(a) (1)
(b) (2)
(c) (3)
(d) all of them are affected by the same magnitude of force

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16 ✳ The following figures illustrate three wires, each of them carries a direct electric current while being in a uniform magnetic field:

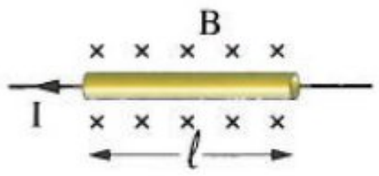


Figure (1)

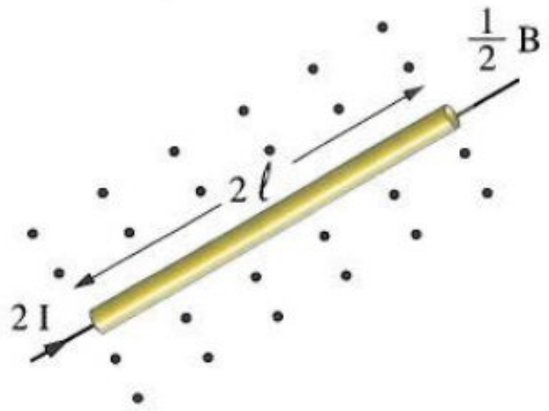


Figure (2)



Figure (3)

Which of the following represents the relation between the magnitudes of the magnetic forces which act on the wires?

(a) $F_1 > F_2 > F_3$ (b) $F_2 > F_1 > F_3$ (c) $F_3 > F_2 > F_1$ (d) $F_3 > F_1 > F_2$

17 A straight wire carrying an electric current of intensity I is placed in three uniform magnetic fields of the same magnetic flux density but in different directions, as shown in the figures below.

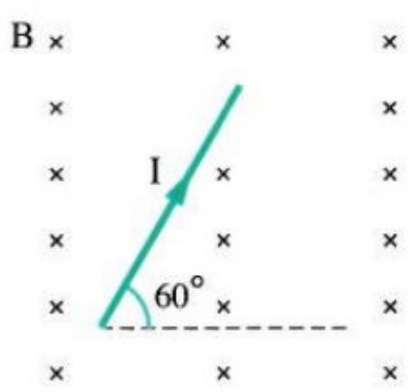


Figure (1)

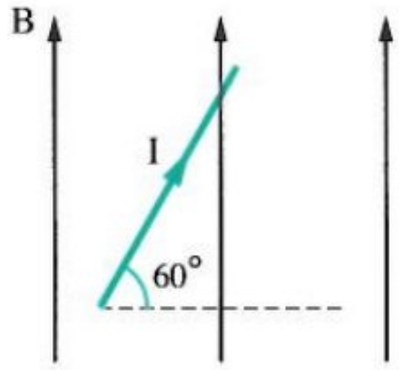


Figure (2)

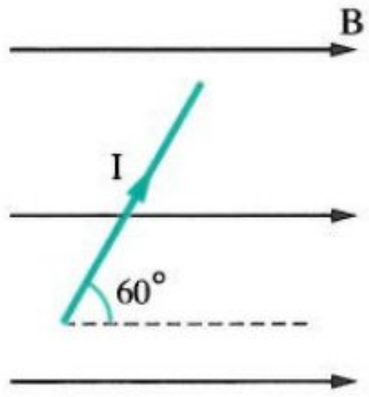
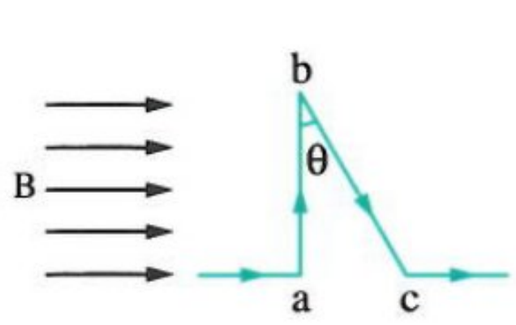


Figure (3)

Then, the correct order of the figures in terms of the magnitude of the magnetic force acting on the wire is ..... .

(a) (1) = (2) = (3) (b) (1) > (2) > (3) (c) (1) > (3) > (2) (d) (2) > (3) > (1)

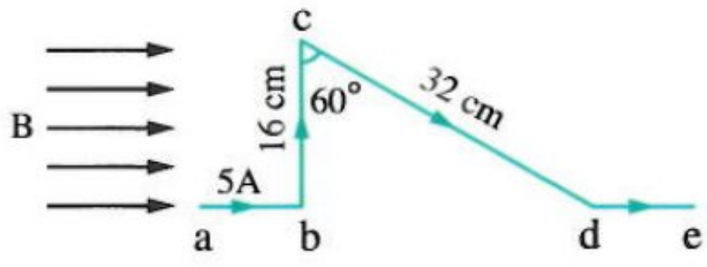
18 ✳ In the opposite figure, if the magnetic force acting on segment ab is F, then the magnitude of the magnetic force acting on segment bc ..... .



(a) is less than F (b) is greater than F

(c) equals F (d) equals F sin θ

19 ✳ In the opposite figure, if the passing electric current intensity in the wire is 5 A and the magnetic flux density is 0.15 T, so:



(i) The magnetic force which affects segment ab of the wire due to this flux equals ..... .

(a) 0.24 N (b) 0.12 N (c) 0.06 N (d) 0

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(ii) The magnetic force which affects segment bc of the wire due to this flux equals ..... .

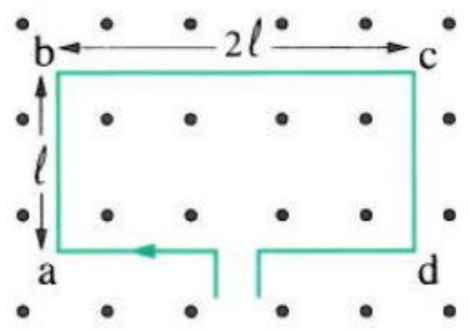
(a) 0.36 N (b) 0.25 N (c) 0.12 N (d) 0

(iii) The magnetic force which affects segment cd of the wire due to this flux equals ..... .

(a) 0.45 N (b) 0.37 N (c) 0.23 N (d) 0.12 N

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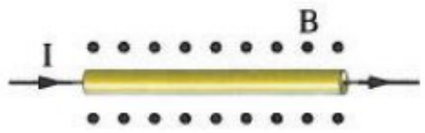
20 The opposite figure shows a rectangular current-carrying coil abcd that is placed perpendicular to a magnetic field, so the ratio between the magnetic force that affects segment ab and the magnetic force that affects segment bc $\left(\frac{F_{ab}}{F_{bc}}\right)$ is ..... .



(a) greater than one (b) less than one (c) equal to one (d) equal to zero

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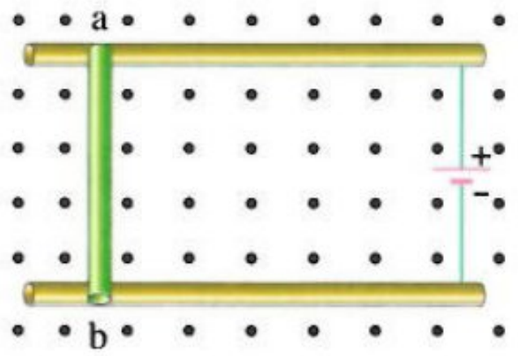
21 A wire of weight F is suspended horizontally parallel to the ground surface and perpendicular to a magnetic flux of density B as shown in the opposite figure, then when an electric current passes through the wire, the wire gets affected by a magnetic force of magnitude 2 F, so the resultant force on the wire equals ..... .



(a) F (b) 2 F (c) $\sqrt{5}$ F (d) 3 F

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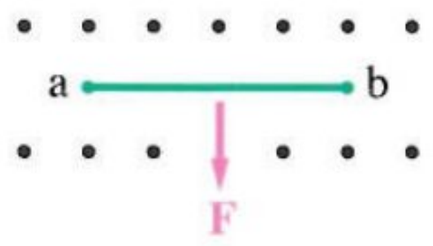
22 The opposite figure shows a cylindrical metallic rod ab that has a length of 20 cm, a resistance of 2 Ω and a mass of 400 g. The rod can slide on two frictionless, parallel rails of copper that have negligible resistances. A 6 V battery has been connected between the terminals of the two copper rails while a magnetic field of flux density 0.1 T is being directed perpendicular to the rod ab. What is the acceleration of the rod at the beginning of its motion? (Where: F = ma)



(a) 3 m/s^2 (b) 1.5 m/s^2 (c) 0.15 m/s^2 (d) 0.015 m/s^2

---

23 A straight wire ab of length 1.5 m that is placed perpendicularly in a magnetic field of 0.2 T. If the magnetic force that affects the wire is 2.4 N in the direction shown in the figure, then the value of the current I and its direction in the wire are ..... .



(a) 16 A, from a to b (b) 8 A, from a to b

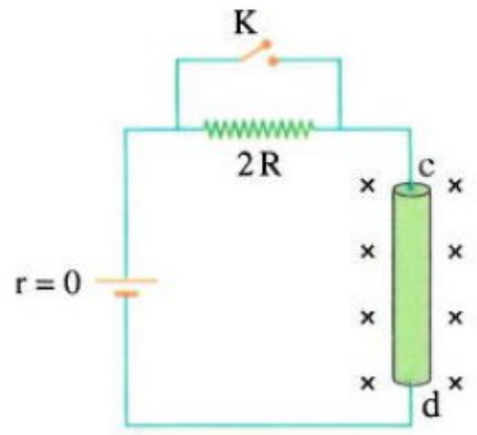
(c) 16 A, from b to a (d) 8 A, from b to a

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24 A rod (cd) of electric resistance R, is placed perpendicular to a uniform external magnetic field (B) and it is connected to an electric circuit as shown in the opposite diagram. If the magnetic force acting on the rod due to the external field is equal to F_o, then the magnitude of this force when the switch K is closed becomes ..... .



(a) $\frac{1}{3} F_o$ (b) $\frac{1}{2} F_o$ (c) $2 F_o$ (d) $3 F_o$

25 Two straight wires x, y made of the same material, having the same cross-sectional area and the length of wire x is greater than the length of wire y, they are placed perpendicular to a uniform magnetic field of flux density B each at a time. If the terminals of each wire are connected to a potential difference V, which of the two wires will be affected by a greater magnetic force?

(a) Wire (x). (b) Wire (y).

(c) Both are affected by the same magnitude of force.

(d) Neither of the wires is affected by a force.

26 ✱ A metallic straight wire of length ℓ, whose cross-sectional area is 10 mm^2 and the resistivity of its material is 2.8×10^{-8} Ω.m, is placed perpendicular to a magnetic field of flux density 10^{-3} tesla. The wire gets connected to a battery of emf 3 V whose internal resistance is negligible, so:

(i) The magnitude of the magnetic force which affects the wire equals ..... .

(a) 0.53 N (b) 0.85 N (c) 1.07 N (d) 2.13 N

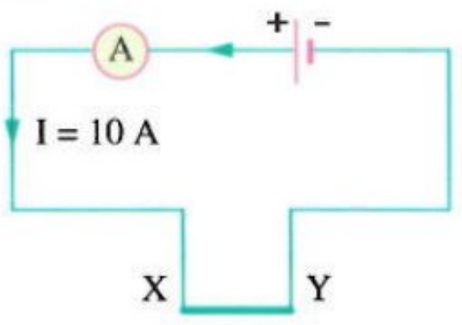
(ii) The magnitude of the magnetic force which affects the wire if its diameter is doubled equals ..... .

(a) 8.36 N (b) 4.28 N (c) 2.15 N (d) 1.07 N

27 A metal straight wire carries an electric current I, so a magnetic flux of density 10^{-5} Wb/m^2 is produced at a point of 10 cm apart from it. If the wire is placed inclined by an angle of 30° in a magnetic flux of density 1.5 Wb/m^2, the magnetic force affecting unit length of the wire equals ..... .

(a) 2.5 N/m (b) 5 N/m (c) 3.75 N/m (d) 10 N/m

28 ✱ Aluminum wire XY of cross-sectional area 0.1 cm^2 is suspended horizontally while its terminals are touching the ends of an electric circuit as shown in the opposite figure, hence the density and the direction of the magnetic flux which keeps the wire suspended without any external support are ..... . (Take: $\rho_{Al} = 2700$ kg/m^3, g = 10 m/s^2)

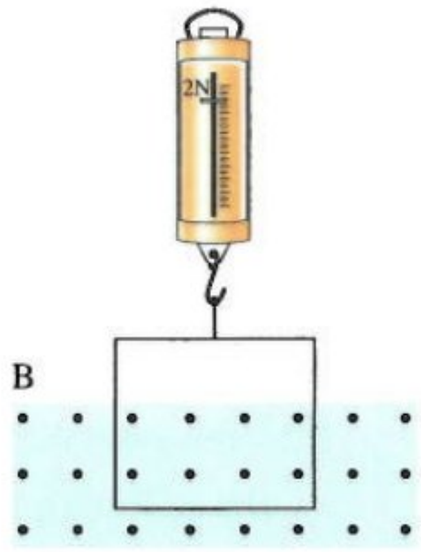


(a) 27×10^{-3} T, perpendicular out of the page (b) 35×10^{-3} T, perpendicular out of the page

(c) 27×10^{-3} T, perpendicular into the page (d) 35×10^{-3} T, perpendicular into the page

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29 A metallic square frame, its side length is 20 cm is suspended to a spring scale whose reading was 2 N, the lower surface of the frame is affected by a magnetic field of a uniform magnetic flux density B as shown in the opposite figure, if an electric current of intensity 20 A passes through the frame in the anti-clockwise direction, then the reading of the scale became 2.2 N, then:



(i) The magnetic flux density (B) equals ..... .

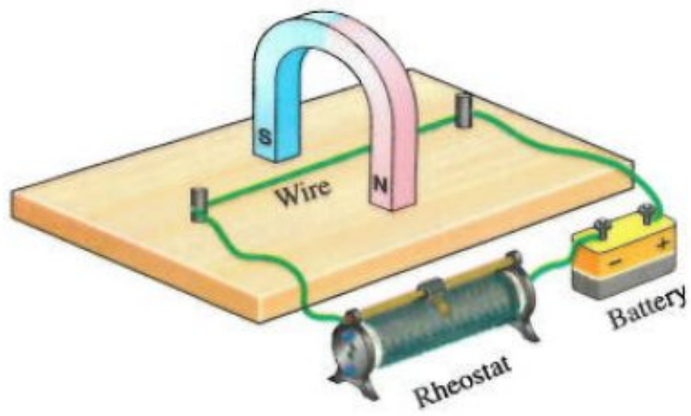
(a) 0.1 T (b) 0.05 T

(c) 0.04 T (d) 0.02 T

(ii) The reading of the spring when the direction of the field is reversed equals ..... .

(a) 2.2 N (b) 2 N (c) 1.8 N (d) 0.2 N

30 * A straight wire of weight w is placed horizontally perpendicular to a magnetic field, as shown in the opposite diagram. When an electric current of intensity I passed through the wire, the net force acting on it was equal to 3 w and directed vertically downward. Then the magnitude and the direction of the net force acting on the wire if:



(i) The direction of the magnetic field is reversed?

	The magnitude of the net force	The direction of the net force
(a)	w	Vertically upward
(b)	w	Vertically downward
(c)	3 w	Vertically upward
(d)	3 w	Vertically downward

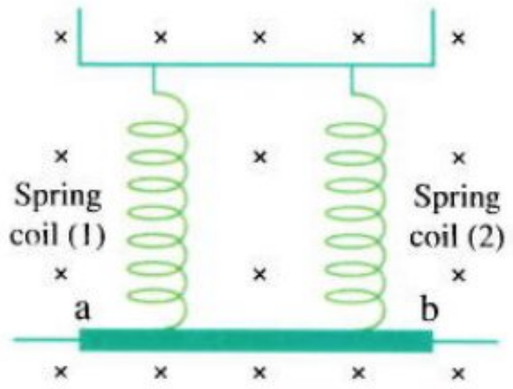
(ii) The electric current passing through the wire is decreased to half its intensity?

	The magnitude of the net force	The direction of the net force
(a)	$\frac{3}{2}$ w	Vertically upward
(b)	$\frac{3}{2}$ w	Vertically downward
(c)	2 w	Vertically upward
(d)	2 w	Vertically downward

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31 ✳ Metallic rod ab of length 0.4 m and mass 50 g is suspended by two similar insulated spring coils of negligible masses and it is affected by a magnetic field of flux density 0.2 T as in the opposite figure, so:
(Take: The acceleration due to gravity = 10 m/s^2)



(i) The intensity and the direction of the electric current in the rod which make the tension forces in the spring coils equal zero are ..... .

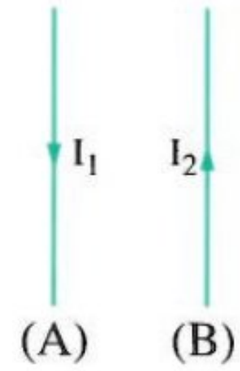
(a) 6.25 A, from a to b
(b) 4.5 A, from a to b
(c) 6.25 A, from b to a
(d) 4.5 A, from b to a

(ii) The magnitudes of tension forces in each spring coil if only the direction of the current is reversed while its intensity is kept unchanged are ..... .

(a) 0.12 N (b) 0.25 N (c) 0.5 N (d) 0.7 N

The mutual force between wires

32 In the opposite figure, two wires (A) and (B) are carrying two electric currents I_1 and I_2 where $I_2 < I_1$, so, the direction of the acting magnetic force on wire (B) is ..... .

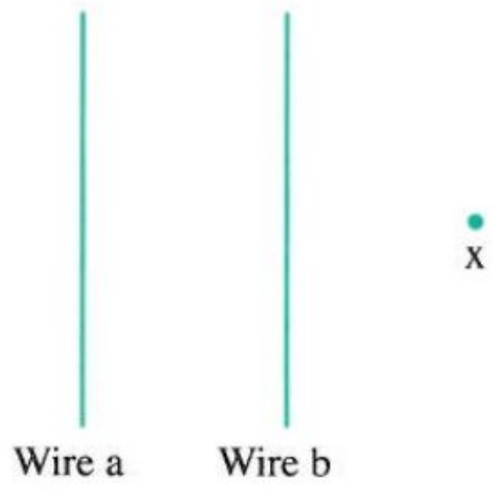


(a) into the page
(b) out of the page
(c) to the right side of the page
(d) to the left side of the page

33 The type of the mutual magnetic force between two current-carrying, parallel wires depends on ..... .

(a) the direction of the current in each of them
(b) the intensity of the current in each of them
(c) the distance between them
(d) their lengths

34 In the opposite figure, two straight parallel wires a, b , each carry an electric current. If the point x shown in the diagram represents the neutral point, then ..... .



(a) a repulsion force arises between two wires
(b) attraction force arises between two wires
(c) no magnetic force arises between two wires
(d) two equal electric currents pass through them

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35 Both of the following figures, 1 and 2, represent the magnetic field generated by the passage of an electric current through two long straight parallel wires placed perpendicular to the plane of the page.

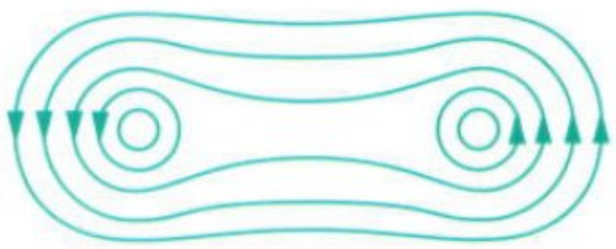


Figure (1)

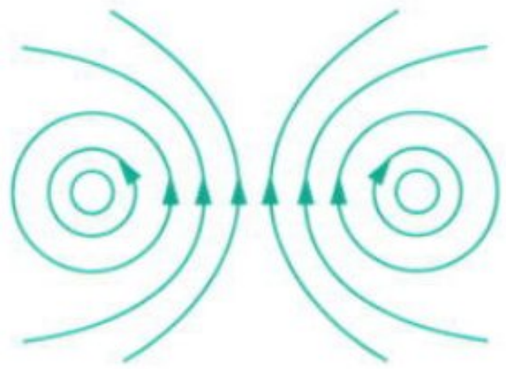


Figure (2)

In which of the two figures does the mutual magnetic force between the two wires become an attractive force?

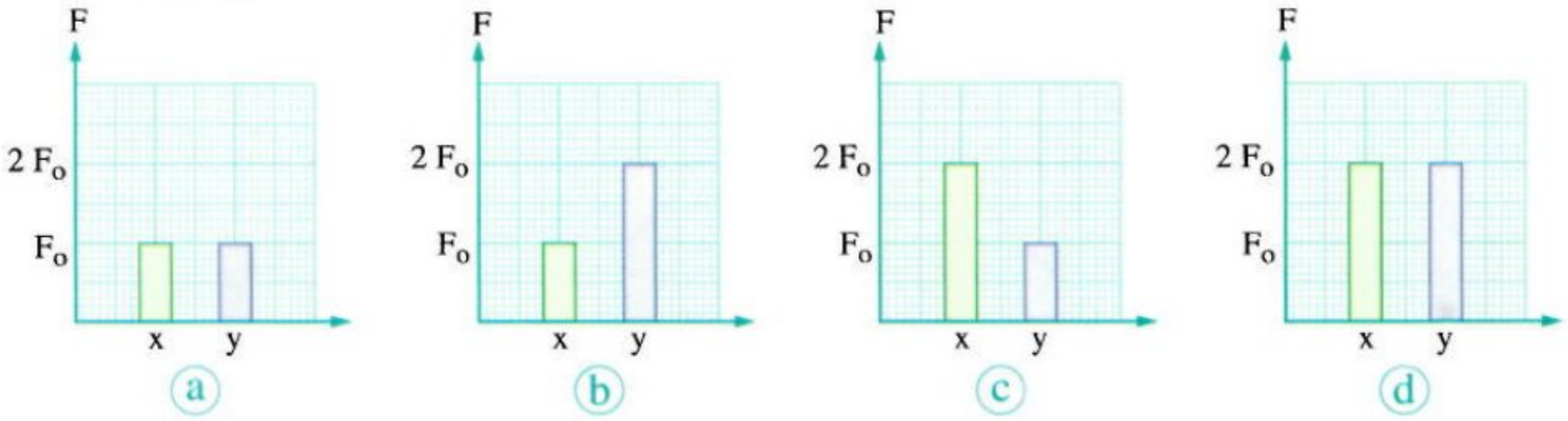
(a) In figure (1).
(b) In figure (2).
(c) In both figures.
(d) Neither of the figures.

36 In the opposite figure, the ratio between the magnetic force acting on wire x to that acting on wire y is ..... .

(a) $\frac{1}{1}$
(b) $\frac{2}{1}$
(c) $\frac{1}{2}$
(d) $\frac{1}{4}$

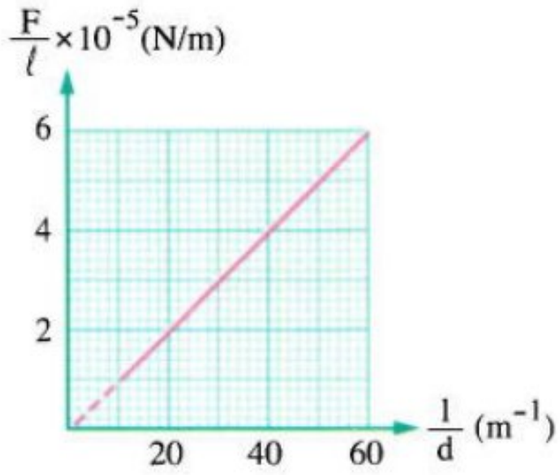


37 Two parallel straight wires x, y carry an electric current of intensity I and the magnetic force acting on each of them was F_o, if the electric current through wire x is increased to double, which of the following figures represents the magnetic force acting on each of the two wires x, y?



38 The opposite figure represents the relation between the mutual magnetic force per unit length between two long parallel wires $(\frac{F}{\ell})$ and the reciprocal of the vertical distance between the two wires $(\frac{1}{d})$. If the current intensities that pass in the two wires have the same value, so the current intensity through any of them equals ..... .

(a) 1 A
(b) 1.51 A
(c) 2.24 A
(d) 3 A



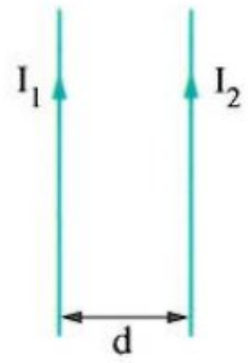
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39 Two long parallel wires are separated by a distance d, both of them carry a current of intensity 10 A in the same direction, if the mutual magnetic force per unit length between them is 2×10^{-4} N/m, so distance d equals ..... .

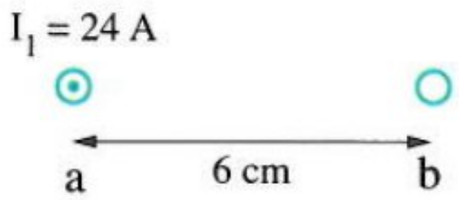
(a) 5 cm (b) 10 cm (c) 15 cm (d) 20 cm

40 In the opposite figure, two very long parallel wires are carrying electric currents where the mutual magnetic force between them is 0.01 N, so if the intensity of one of the two currents is doubled and the distance between them is halved, then the mutual magnetic force between them becomes ..... .



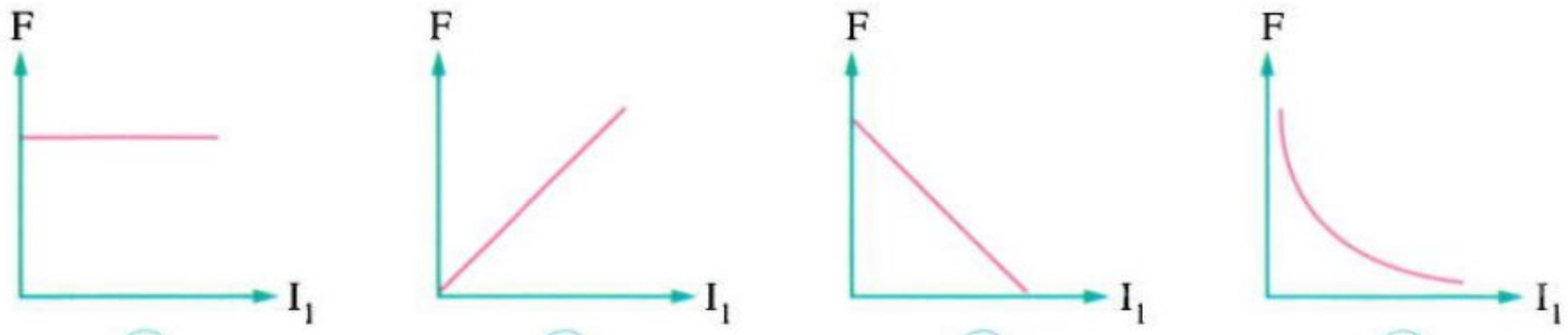
(a) 0.04 N (b) 0.02 N (c) 0.01 N (d) 0.005 N

41 The opposite figure shows two straight parallel wires a and b which are separated by a distance 6 cm. An electric current of intensity 24 A flows through wire a out of the plane of the page. If wire a is affecting wire b by an attraction force whose magnitude per unit length equals 8.8×10^{-4} N/m, so the magnitude and the direction of the current which passes through wire b are ..... respectively.

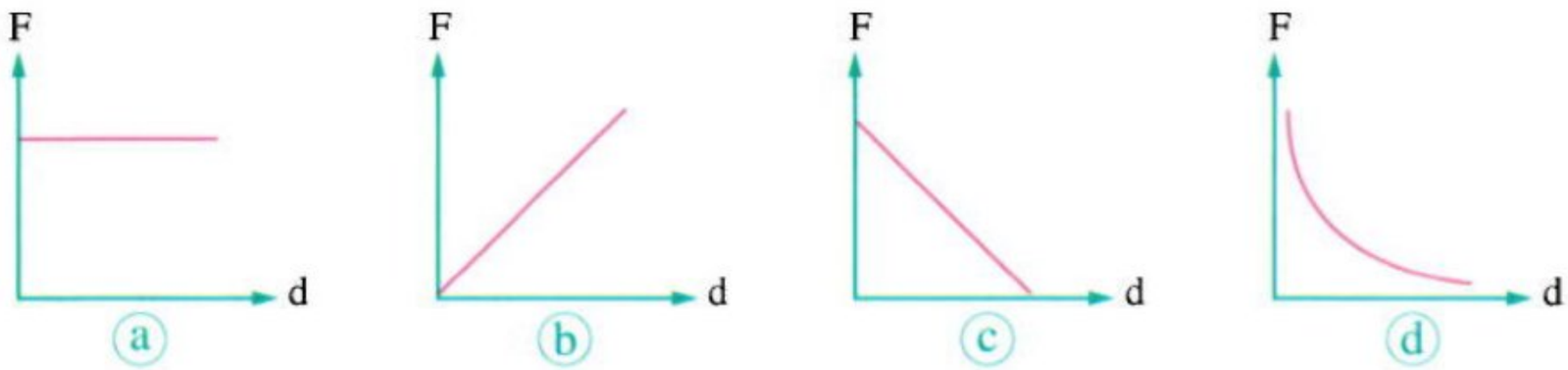


(a) 11 A, out of the page (b) 12.5 A, out of the page
(c) 11 A, into the page (d) 12.5 A, into the page

42 Which of the following graphs represents the relation between the mutual magnetic force (F) between two parallel current-carrying wires and the intensity of the current (I_1) which is passing in one of the two wires while holding the other factors constant?

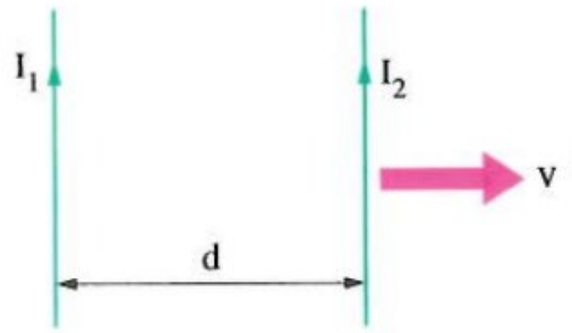


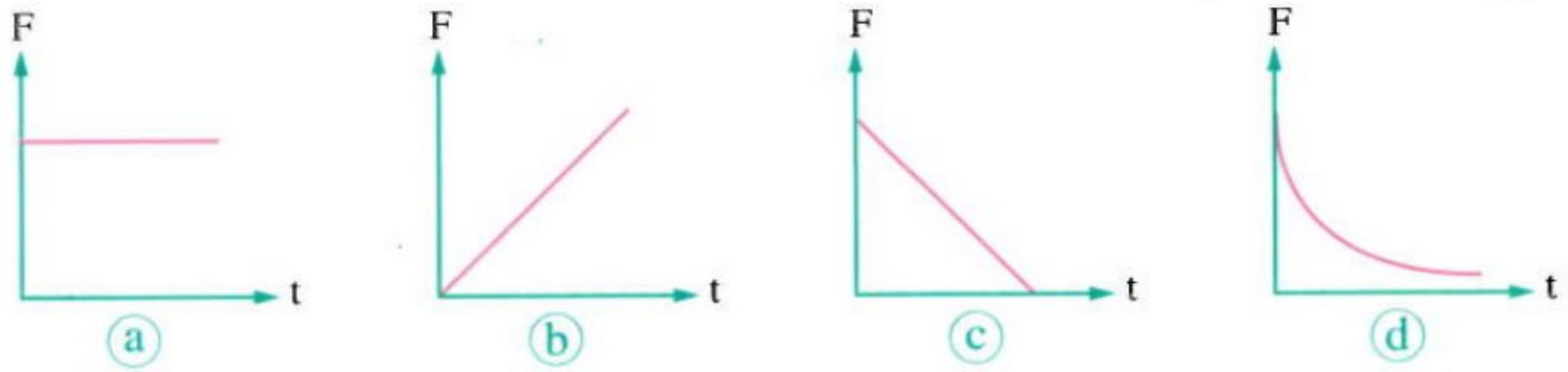
43 Which of the following graphs represents the variation of the mutual force (F) between two parallel current-carrying wires and the distance (d) between them?



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44 In the shown figure, if the second wire is moving to the east with a uniform speed (v), which of the following graphs represents the variation of the mutual force (F) between the two wires with time (t)?





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45 ✱ A very long straight wire carries an electric current of intensity 5 A:

(Where : $\frac{\mu}{2\pi} = 2 \times 10^{-7}$ Wb/A.m)

(i) So, the formed magnetic flux density, due to the flow of the electric current in the wire, at a point in air which is separated by a perpendicular distance of 10 cm from the wire equals ..... .

(a) 1×10^{-5} T (b) 2×10^{-5} T (c) 3×10^{-5} T (d) 4×10^{-5} T

(ii) If another wire of length 50 cm is placed at a perpendicular distance of 10 cm parallel to the first wire while carrying an electric current of intensity 2 A, the force affecting this wire due to the field of the first wire equals ..... .

(a) 2.5×10^{-5} N (b) 1×10^{-5} N (c) 0.5×10^{-4} N (d) 0.25×10^{-4} N

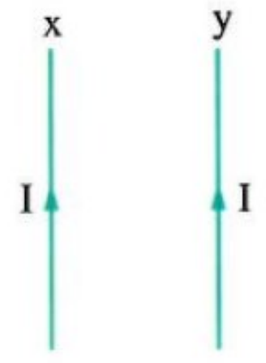
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46 ✱ Two straight, parallel wires which are separated by a distance of 2 m in air carry electric currents in the same direction. If the magnetic flux density has vanished at a point in the middle of the distance between the two wires and the acting force on one meter of any of them is 4×10^{-5} N, the intensities of the passing currents through each wire equal ..... .

(a) 10 A, 20 A (b) 15 A, 3 A (c) 10 A, 10 A (d) 20 A, 20 A

---

47 In the opposite figure, two straight, parallel wires x, y were carrying electric currents of the same intensity I and in the same direction, hence a magnetic force (F) is generated between them. When the current of wire x is increased by 4 A, the mutual force between the two wires is doubled. So, the value of I was equal to ..... .



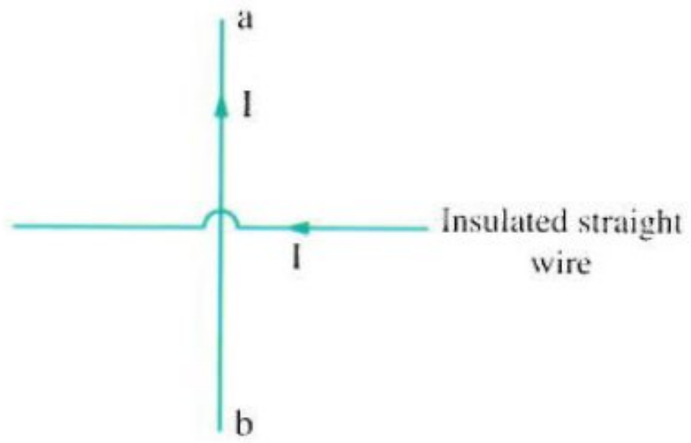
(a) 2 A (b) 4 A (c) 6 A (d) 8 A

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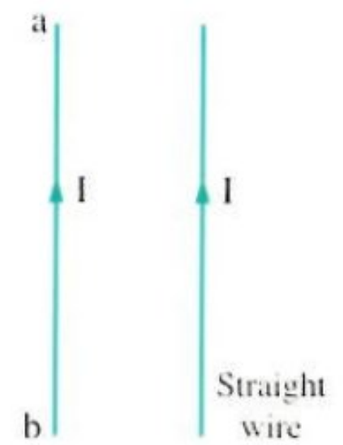
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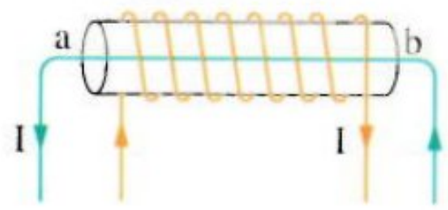
48 In which of the following cases is the insulated straight wire ab not affected by a net magnetic force?



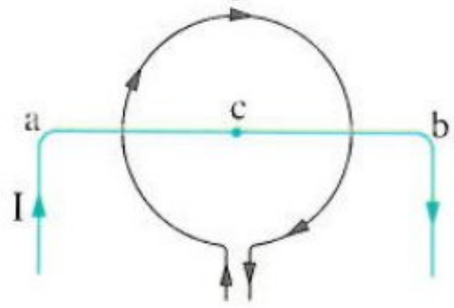
(a)



(b)

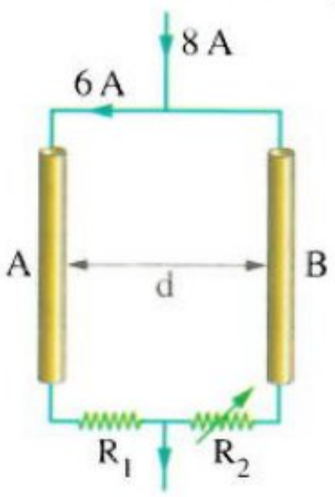


(c)

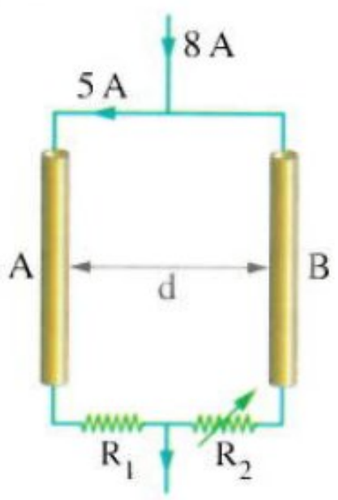


(d)

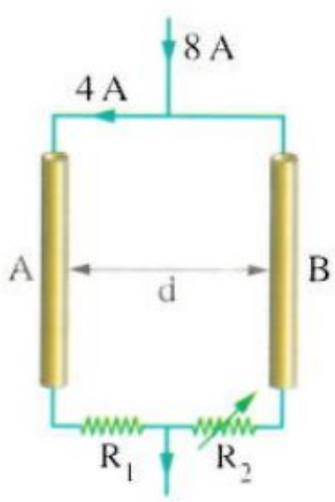
49 Two identical parallel straight wires A, B, the distance between them is d, if they are added to an electric circuit and the value of the variable resistance R_2 is changed, the intensity of the electric current passes through each of them is changed as shown in the following figures. In any of these figures, the magnitude of the mutual force between the two wires is the greatest?



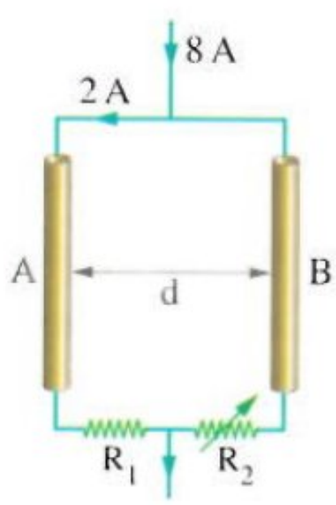
(a)



(b)

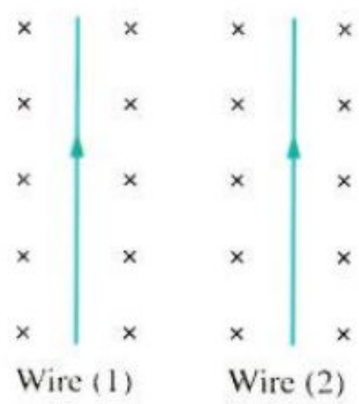


(c)



(d)

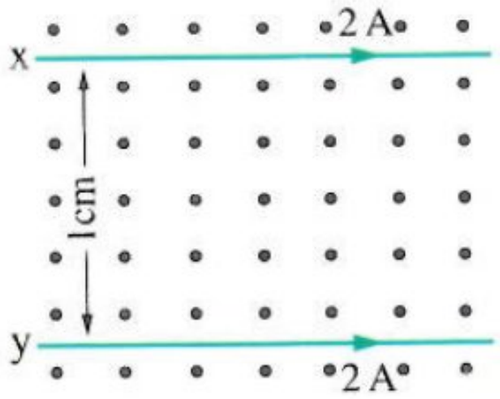
50 Two long parallel wires (1), (2), each carrying an electric current of intensity I , are placed perpendicular to an external uniform magnetic field, as shown in the opposite diagram. If the net magnetic force acting on unit length for each wire of the two wires is F_1 and F_2, respectively, which of the following relation is correct?



(a) $F_1 = F_2 = 0$ (b) $F_1 > F_2$ (c) $F_1 = F_2 \neq 0$ (d) $F_1 < F_2$

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51 ✳ Two long parallel wires x, y, each carrying an electric current of intensity 2A . are placed in an external uniform magnetic field of density 2×10^{-5} T and its direction is perpendicular to each wire. What is the total magnetic force acting per unit length for:



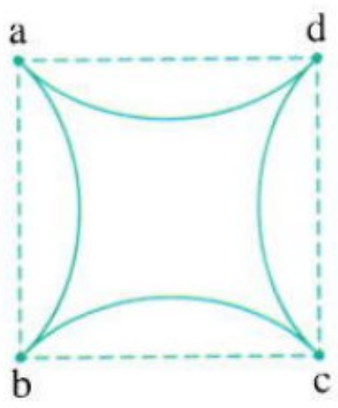
(i) wire x?

(a) 1.2×10^{-5} N/m (b) 4×10^{-5} N/m (c) 8×10^{-5} N/m (d) 1.2×10^{-5} N/m

(ii) wire y?

(a) 1.2×10^{-5} N/m (b) 4×10^{-5} N/m (c) 8×10^{-5} N/m (d) 1.2×10^{-4} N/m

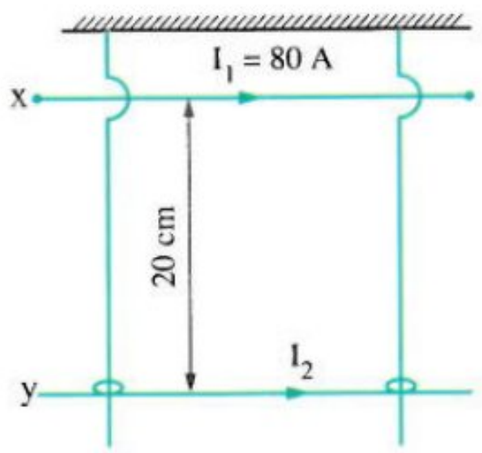
52 An un-insulated elastic metal wire is shaped into a square abcd. Which two points on the frame must be connected to a DC source to take the position shown in the opposite figure?



(a) b, a (b) d, a

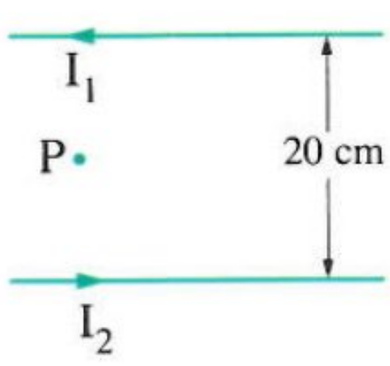
(c) c, d (d) b, d

53 ✳ The opposite figure represents two straight horizontal wires (x, y) in the same plane where wire x is fixed horizontally while carrying a current of intensity 80 A at a distance 20 cm from wire y that can move up and down. If the mass of one meter of the wire y is 0.12 g/m, then the current intensity (I_2) that has to pass in this wire so it remains suspended and doesn't fall under the effect of gravity equals ..... . (Take: g = 10 m/s^2)



(a) 15 A (b) 20 A (c) 30 A (d) 40 A

54 ✳ Two parallel, straight wires, each of them has a length of 50 cm, are separated by a distance of 20 cm in air. The first wire carries a current of intensity I_1 and the second carries a current of intensity I_2 = 10 A in the directions that are shown in the opposite figure. If the total magnetic flux density at point P in the middle of the distance between the two wires is 6×10^{-5} T, the mutual magnetic force between them equals ..... .



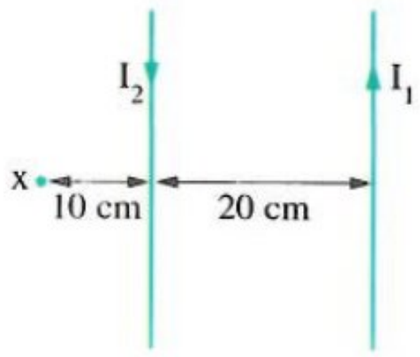
(a) 10^{-2} N (b) 10^{-3} N (c) 10^{-4} N (d) 10^{-5} N

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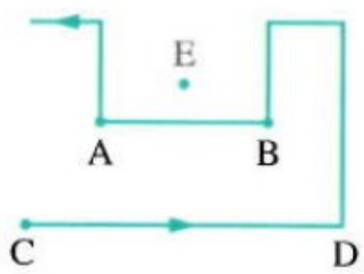
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55 ✱ In the opposite figure, if point x is the neutral position and the magnetic force affecting one meter of any of the two wires is 12×10^{-6} N, the intensities of currents I_1 and I_2 respectively are ..... .



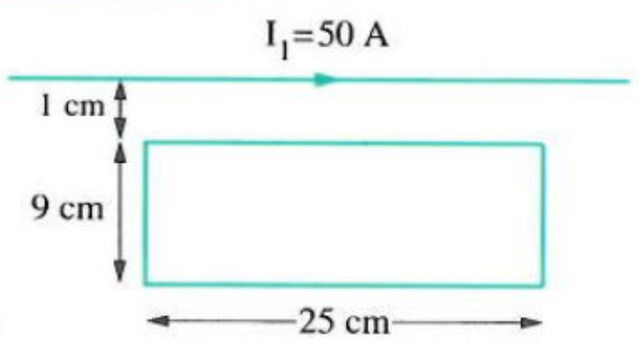
(a) 3 A, 1 A (b) 1 A, 1 A
(c) 6 A, 4 A (d) 6 A, 2 A

56 ✱ In the opposite figure, two horizontal wire segments AB and CD are in one vertical plane carrying a current of intensity 50 A. Wire segment AB which is 1 m long and has a mass of 5 g is free to move vertically, while wire segment CD is fixed, so by neglecting the magnetic flux due to the vertical segments of the wires, the required distance between the two wire segments to hold segment AB balanced equals ..... . (Take: g = 10 m/s^2)



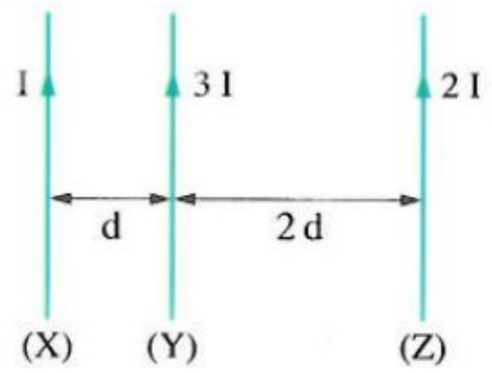
(a) 0.23 m (b) 0.15 m (c) 0.02 m (d) 0.01 m

57 ✱ The opposite figure represents a frame of dimensions 25 cm, 9 cm where its mass is 4.5 g. A long straight horizontal wire that carries an electric current of 50 A is placed above it, so the intensity and the direction of the current that should be passing through the frame to remain vertically suspended in air are ..... . (Take: g = 10 m/s^2)



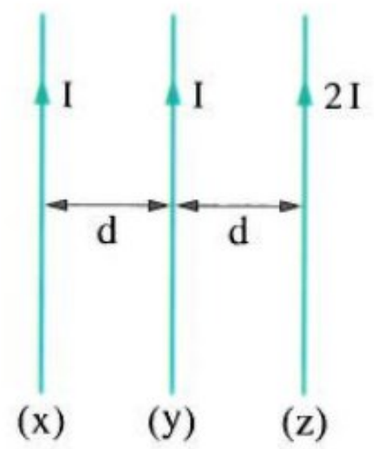
(a) 200 A, anti-clockwise (b) 200 A, clockwise
(c) 100 A, anti-clockwise (d) 100 A, clockwise

58 In the opposite figure, three long parallel wires carry electric currents, so which of these wires are not affected by a net magnetic force?



(a) Z (b) Y
(c) X (d) Both X and Z

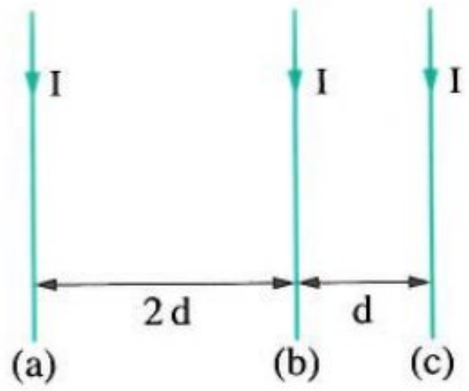
59 Three long straight parallel wires x , y and z carry currents in the same direction of intensities I , I and 2I respectively as shown in the opposite figure. Which of the wires is affected by the greatest net magnetic force?



(a) Wire x
(b) Wire y
(c) Wire z
(d) The three wires are affected by the same magnitude of force.

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60 In the opposite figure, three long parallel wires (a), (b) and (c) are in the plane of the page carrying electric currents of the same intensity in the directions shown in the figure, so the direction of the net magnetic force acting on wire (b) is ..... .



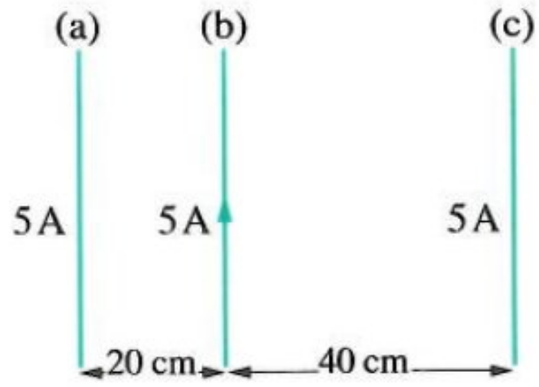
(a) perpendicular into the plane of the page

(b) perpendicular out of the plane of the page

(c) in the plane of the page towards the right side

(d) in the plane of the page towards the left side

61 * In the opposite figure, three parallel wires each carries electric current of intensity 5 A. They are placed in the same plane, hence the force affecting one meter of wire (b), when the two electric currents in the two other wires (a) and (c):



(i) are in the same direction, equals ..... .

(a) 50×10^{-6} N (b) 37.5×10^{-6} N

(c) 25×10^{-6} N (d) 12.5×10^{-6} N

(ii) are in opposite directions, equals ..... .

(a) 75×10^{-6} N (b) 37.5×10^{-6} N (c) 19×10^{-6} N (d) 14×10^{-6} N

62 * Two straight, parallel wires are separated by a distance of 10 cm. A current of intensity 2 A passes in one of them while 3 A in the other and in the same direction, so:

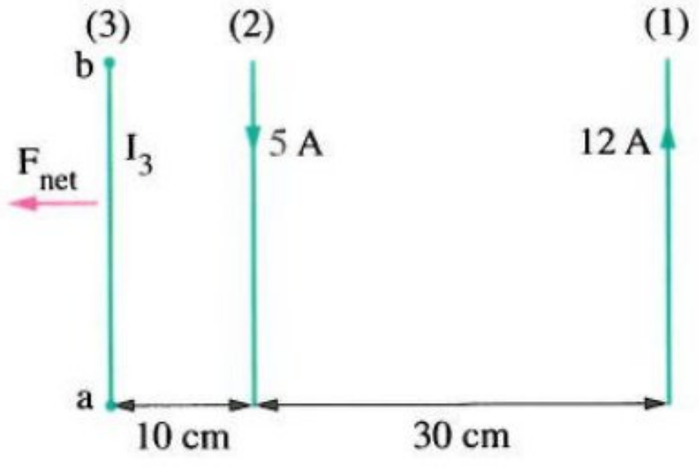
(i) The distance of the neutral point from the wire that carries the 2 A current equals ..... .

(a) 2.5 cm (b) 4 cm (c) 6.5 cm (d) 8 cm

(ii) If the direction of the current in one of the wires is reversed and a third wire of length 10 cm that carries a current of 5 A is placed parallel to them and in the same plane at the neutral point of the previous part, the force affecting this wire will equal ..... .

(a) 7.3×10^{-5} N (b) 5×10^{-5} N (c) 2.5×10^{-5} N (d) 10^{-5} N

63 The opposite figure represents three long straight wires which are placed in the plane of the page, each of them carries an electric current in the direction shown in the figure. If the resultant magnetic force acting on unit length of wire (3) is 3×10^{-6} N/m and its direction is parallel to the plane of the page to the left side, then the intensity and the direction of the current in wire (3) is ..... .



(a) 0.75 A, from b to a

(b) 0.75 A, from a to b

(c) 5 A, from b to a

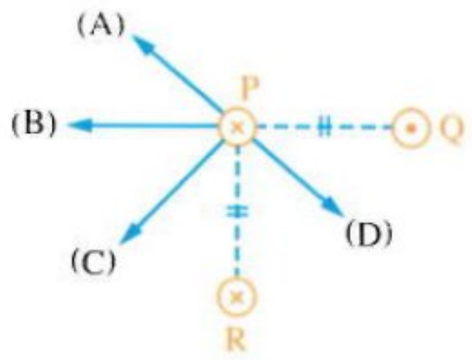
(d) 5 A, from a to b

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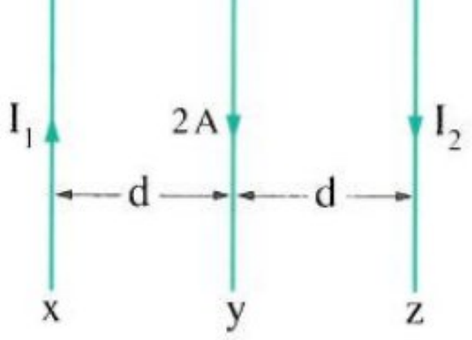
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64 The opposite figure shows three long wires P, Q and R which are placed normal to the page. If all of them carry currents of the same intensity, however, the directions of the currents in P and R are into the page and the direction of the current in Q is out of the page, so which of the shown arrows (A, B, C, or D) represents the direction of the resultant magnetic force on wire P?



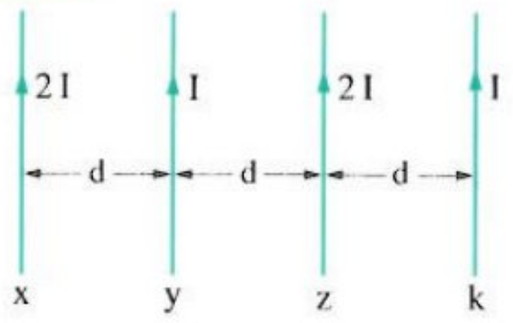
(a) A (b) B (c) C (d) D

65 ✳ In the opposite figure, three parallel, straight current-carrying wires are in the same plane, if you know that the magnetic force that affects a unit length of wire y is F and when the current of wire x is reversed, the force that affects a unit length of wire y becomes $\frac{1}{2}$ F. Given that $I_2 < I_1$, so the ratio between the two currents $(\frac{I_1}{I_2})$ equals ..... .



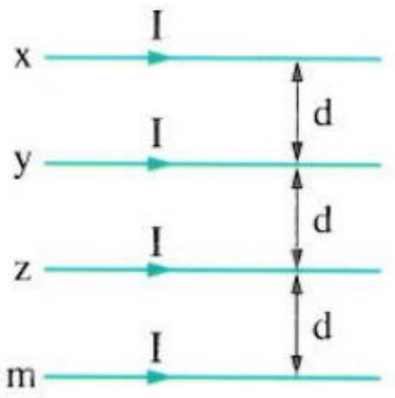
(a) $\frac{2}{1}$ (b) $\frac{3}{1}$ (c) $\frac{4}{1}$ (d) $\frac{5}{4}$

66 ✳ Four straight parallel wires x, y, z, k, each of them carries an electric current as shown in the opposite figure, then the magnitude of the magnetic force for a unit length of wire z is equal to ..... .



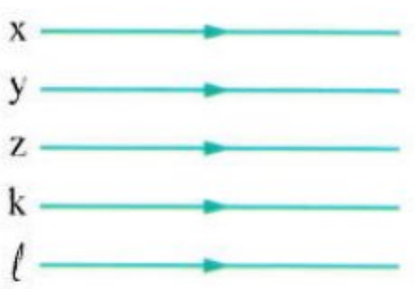
(a) $\frac{1}{2} \times \frac{\mu I^2}{\pi d}$ (b) $\frac{\mu I^2}{\pi d}$ (c) $2 \times \frac{\mu I^2}{\pi d}$ (d) $4 \times \frac{\mu I^2}{\pi d}$

67 The opposite figure shows four identical parallel wires x, y, z and m carry an electric current of intensity I in the same direction, if the resultant of the magnetic force acting on wire z equals F, then the resultant of the magnetic force acting on wire m equals ..... .



(a) $\frac{11}{3}$ F (b) $\frac{11}{6}$ F (c) $\frac{11}{5}$ F (d) $\frac{11}{2}$ F

68 Five parallel straight wires x, y, z, k and ℓ are separated by equal distances and each of them carries an electric current of intensity I as shown in the opposite figure, so the correct order for the wires in terms of the value of the magnetic force affecting each of them is .....



(a) $z > x = \ell > y = k$ (b) $y = k > x = \ell > z$
(c) $x = \ell > z > y = k$ (d) $x = \ell > y = k > z$

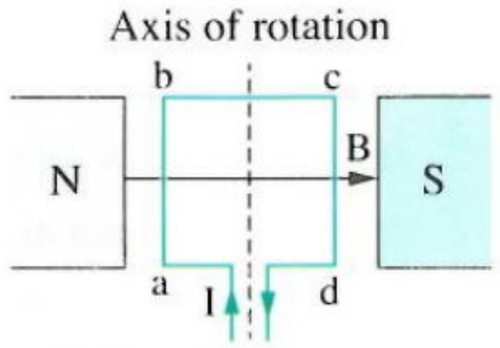
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The magnetic torque

69 The torque (τ) acting on a current-carrying coil that is placed in a uniform magnetic field becomes maximum when the plane of the coil is ..... the direction of the magnetic field.

(a) perpendicular to (b) parallel to
(c) inclined with an angle of 30° to (d) inclined with an angle of 60° to

70 A square coil abcd carries an electric current of intensity I and is placed in magnetic field of flux density B as shown in the opposite figure, if the coil is rotated a quarter cycle from the position shown in the figure, what happens to :



(i) the magnitude of magnetic force affecting side ab?

(a) Increases. (b) Decreases but doesn't vanish.
(c) Decreases till vanishes. (d) Remains unchanged.

(ii) the magnetic force affecting side bc?

(a) Increases. (b) Decreases but doesn't vanish.
(c) Decreases till vanishes. (d) Remains unchanged.

(iii) the magnetic torque affecting the coil?

(a) Increases. (b) Decreases but doesn't vanish.
(c) Decreases till vanishes. (d) Remains unchanged.

71 ✱ A coil consists of 200 turns where each turn has a cross-sectional area of 300 cm^2, if the coil of the motor is positioned parallel to a magnetic field of flux density 0.4 T, it gets affected by a torque of 20 N.m. So, the intensity of the passing current in the coil equals ..... .

(a) 4.37 A (b) 8.33 A (c) 11.53 A (d) 13.98 A

72 A coil of 500 turns, each turn has a cross-sectional area of 0.2 m^2, carries an electric current of intensity 10 A where it is placed in a magnetic flux of density 0.25 tesla, so the torque affecting the coil, when the angle between the normal to the coil and the direction of the field is 30°, equals ..... .

(a) 50 N.m (b) 75 N.m (c) 110 N.m (d) 125 N.m

73 ✱ A rectangular coil of dimensions 20 cm, 10 cm, that consists of 200 turns of wire through which a current of 3 A is passing, is placed in a uniform magnetic field of 0.4 tesla, so the torque affecting the coil when:

(i) it is positioned at 60° to the direction of the magnetic field equals ..... .

(a) 0 (b) 2.4 N.m (c) 3.6 N.m (d) 4.5 N.m

(ii) its plane is positioned perpendicular to the direction of the magnetic field equals .....

(a) 6.2 N.m (b) 4.8 N.m (c) 2.5 N.m (d) 0

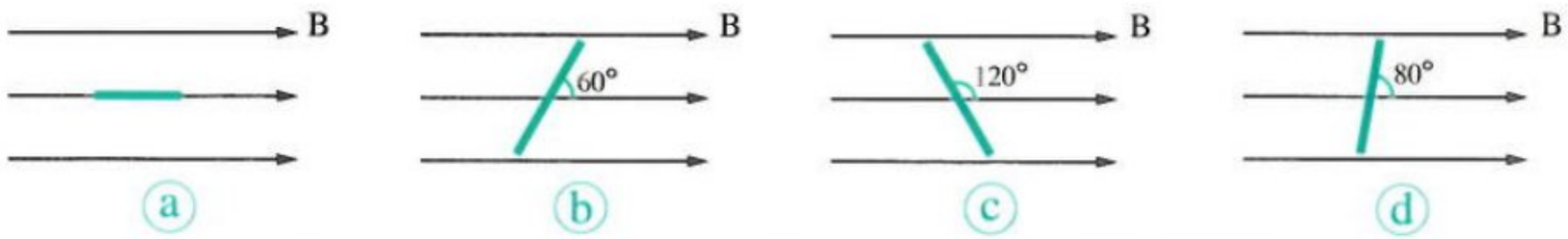
(iii) its plane is positioned parallel to the direction of the magnetic field equals ..... .

(a) 7.4 N.m (b) 0 (c) 4.8 N.m (d) 5.3 N.m

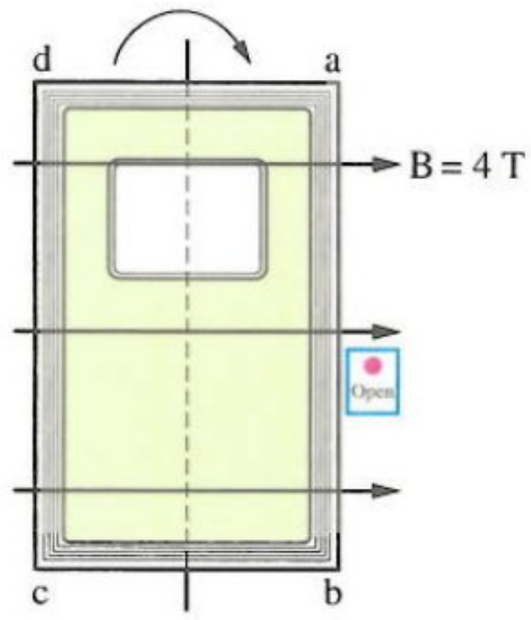
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74 Which of the following positions for a current-carrying coil placed in a uniform magnetic field expresses the smallest value of the torque?

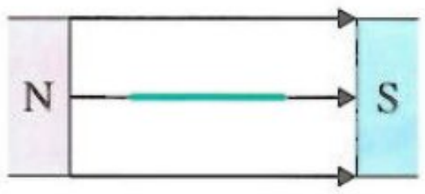


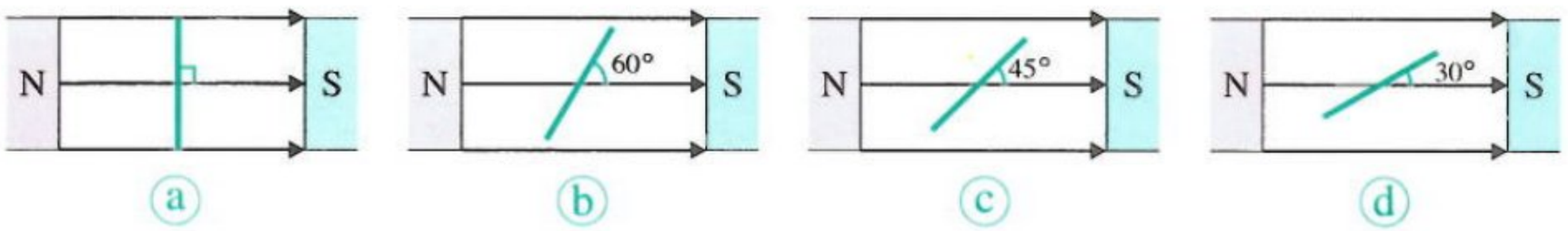
75 An inventor wants to design a door abcd of area 2 m^2 that opens automatically where the door is wrapped by a coil of 50 turns of wires as shown in the figure, such that the door starts rotation about its axis with a torque of 400 N.m from the shown position and the direction of motion of side ab is out of the page, hence the intensity and the direction of the current through side ab of the door have to be ..... .



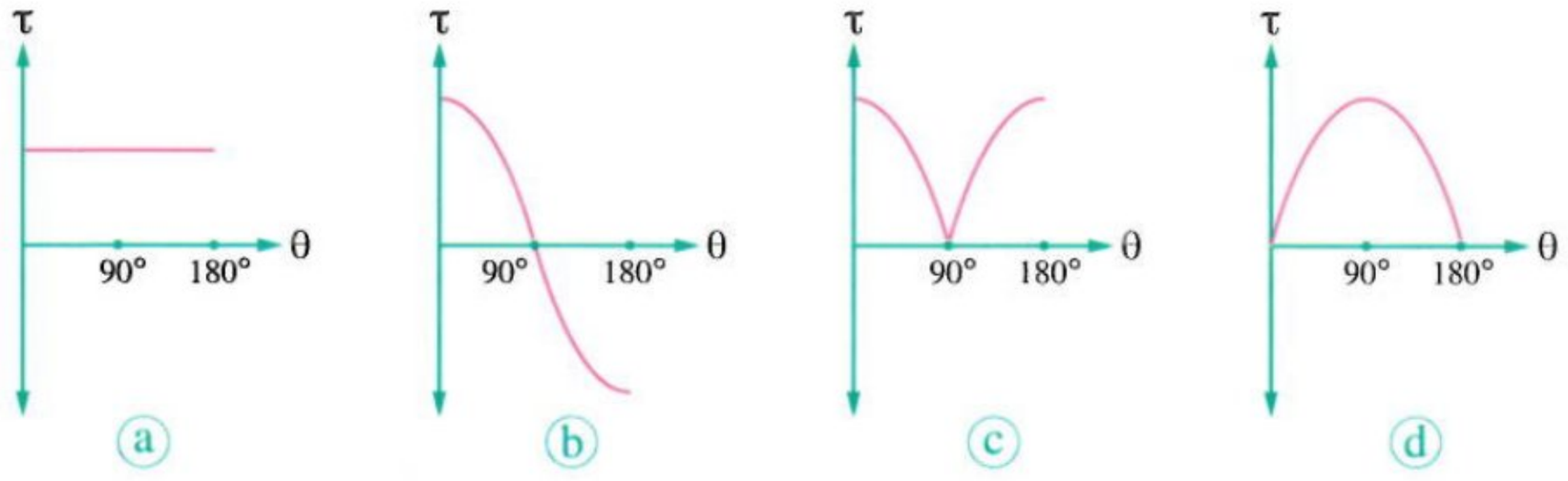
(a) 1 A, from a to b (b) 0.5 A, from a to b
(c) 1 A, from b to a (d) 0.5 A, from b to a

76 The opposite figure shows a lateral view of a rectangular current-carrying coil that is placed in a magnetic field. In which the coil is affected by a torque (τ), so which of the following positions of the coil makes the torque = $\frac{\tau}{2}$?



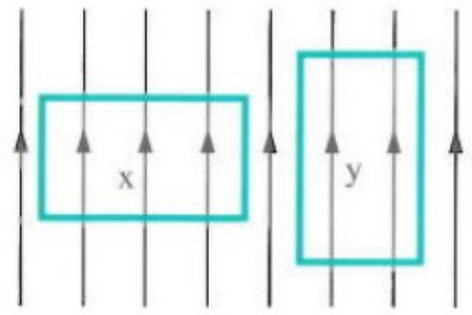


77 Which of the following graphs represents the relation between the torque (τ) on a coil that carries an electric current in a uniform magnetic field and the angle between the coil's plane and the normal to the field (θ) during half cycle from the perpendicular position to the field?



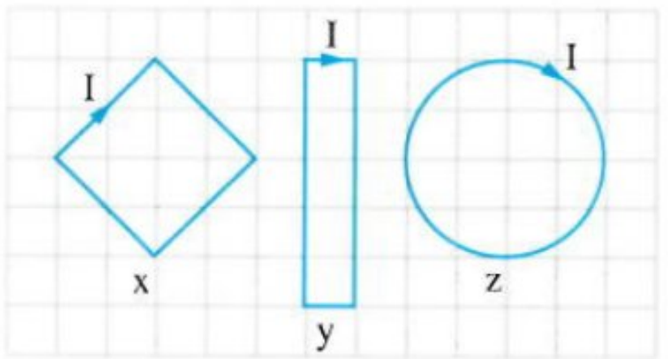
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78 The opposite figure shows two identical frames x and y, whose dimensions are ℓ and 2ℓ. The two frames are placed in a uniform magnetic field, so the ratio of torques which act on each of the frames $(\frac{\tau_x}{\tau_y})$, if both of them carry the same electric current equals ..... .



(a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{3}{1}$ (d) $\frac{1}{1}$

79 Three coils x, y and z, each of them consists of one turn, are different in shape and their dimensions are shown in the opposite graph, if the plane of these coils is parallel to a uniform magnetic field of flux density B and carry an electric current of intensity I, then the correct order for the magnetic torque affecting them is ..... .

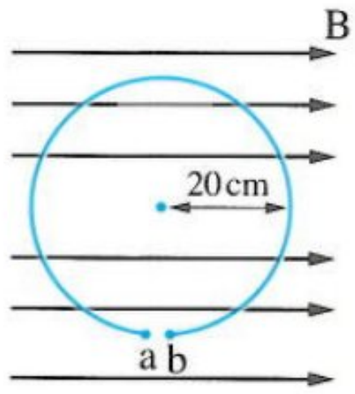


(a) $\tau_x > \tau_y > \tau_z$ (b) $\tau_x > \tau_z > \tau_y$ (c) $\tau_z > \tau_y > \tau_x$ (d) $\tau_z > \tau_x > \tau_y$

80 A rectangular coil has 50 turns of wire through which a magnetic flux of maximum value 0.2 Wb is passing, hence the maximum value of the torque that can affect the coil when it carries a current of intensity 2 A equals ..... .

(a) 20 N.m (b) 40 N.m (c) 60 N.m (d) 80 N.m

81 * In the opposite figure, a metallic ring that is made of a wire of resistance 0.1 Ω nearly forms a complete circle. If a battery of emf 9 V that has a negligible internal resistance is connected between a and b, the torque affecting the ring due to the effect of a uniform magnetic field (B) of flux density 0.4 T that is directed parallel to the plane of the ring equals ..... . (Take: $\pi = 3.14$)



(a) 2.35 N.m (b) 3.75 N.m (c) 3.92 N.m (d) 4.52 N.m

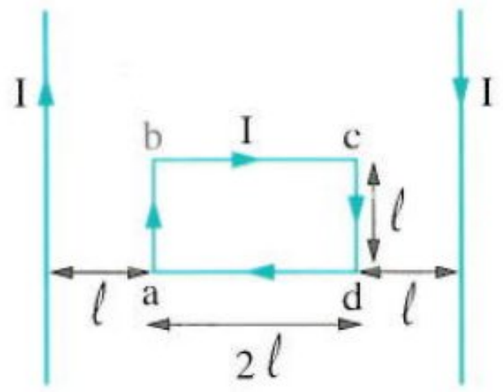
82 * A battery of emf 14 V, whose internal resistance is negligible, is connected to a circular coil of radius 10 cm. If the coil is made of a wire whose material's resistivity is 7×10^{-7} Ω.m and the wire's radius is 1 mm, the torque affecting the coil when its plane is positioned parallel to a magnetic field of flux density 0.5 T equals ..... . (Take: $\pi = 3.14$)

(a) 0.53 N.m (b) 0.61 N.m (c) 0.93 N.m (d) 1.57 N.m

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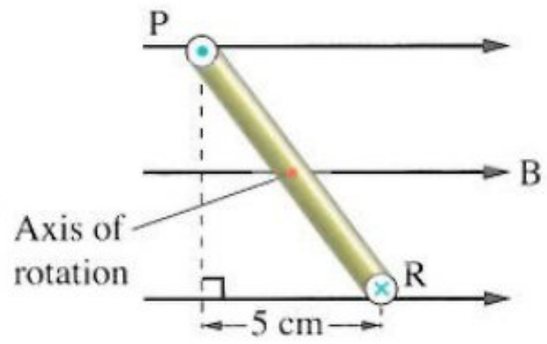
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83 A metallic rectangular frame abcd is placed in the same plane between two long, straight and parallel wires, if currents of equal intensity I are set up in each of them as shown in the opposite figure, the coil (frame) will ..... .



(a) rotate about an axis that is parallel to the two wires
(b) rotate about an axis that is perpendicular to the two wires
(c) move upwards in a direction parallel to the two wires
(d) not be affected by a torque

84 The opposite diagram represents a top view of a rectangular coil that carries an electric current which is directed out of the page at point P and into the page at point R, so if the length of side PR of the coil which is perpendicular to the axis of rotation is 10 cm, how much will be the magnitude of the torque that affects the coil in this position with respect to the maximum value of the torque (τ_o)?



(a) $\sqrt{2}\,\tau_o$ (b) $\frac{1}{\sqrt{2}}\,\tau_o$ (c) $\frac{\sqrt{3}}{2}\,\tau_o$ (d) $\frac{1}{2}\,\tau_o$

85 Two coils x, y, each carries an electric current and placed in a uniform magnetic field B as shown in the two figures (1) and (2) :

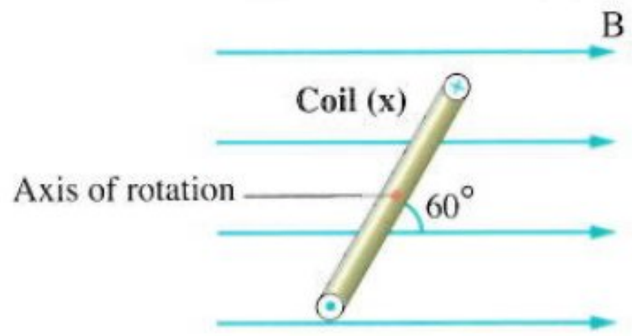


Figure (1)

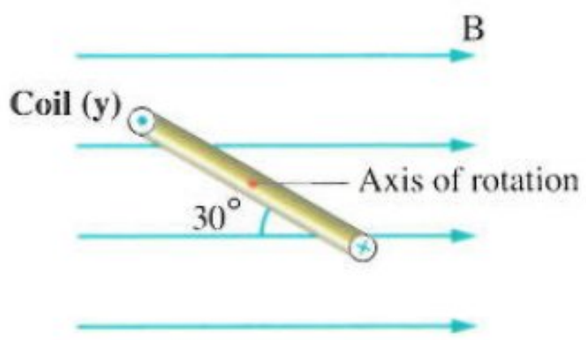
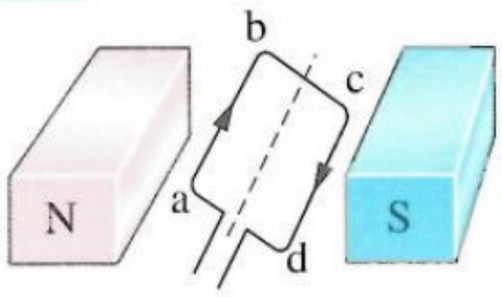


Figure (2)

Which of the two coils, when rotated 60° around the axis shown in the figure, causes the magnetic torque acting on it to vanish?

(a) Coil (x) only. (b) Both of the two coils.
(c) Coil (y) only. (d) Neither of the two coils.

86 The opposite diagram represents a coil abcd of one turn and its dimensions are 5 cm and 10 cm, it carries an electric current and is placed in a uniform magnetic field. If the plane of the coil is inclined at an angle of 60° to the direction of the field and the magnetic force acting on side bc in the given position is 0.5 N, then the value of the magnetic torque acting on the coil in this position is approximately equal to ..... .



(a) 0.01 N.m (b) 0.02 N.m (c) 0.03 N.m (d) 0.05 N.m

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87 The magnetic torque acting on a coil carrying an electric current and rotating in a uniform magnetic field vanishes when the direction of the magnetic dipole moment of the coil is ..... .

(a) parallel to the direction of the field.

(b) perpendicular to the direction of the field.

(c) inclined by angle 45° to the direction of the field.

(d) inclined by angle 135° to the direction of the field.

88 A one-turn square coil that was made of a wire of length 32 cm has been re-wound as a square coil of 4 identical turns, if a current of the same intensity has flowed through the coil in both cases, the magnetic dipole moment of the coil in the first case is ..... of that in the second case.

(a) quadruple (b) double (c) half (d) quarter

89 A rectangular coil that carries an electric current of intensity I is positioned perpendicular to a magnetic flux of density B, so the magnetic dipole moment of the coil increases when ..... .

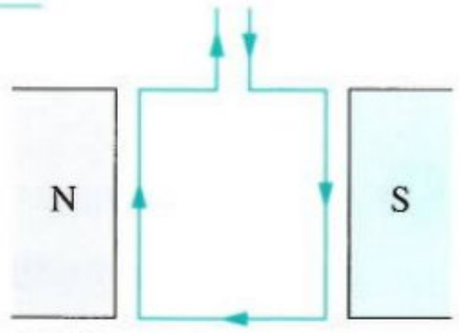
(a) the plane of the coil becomes parallel to the magnetic flux

(b) the magnetic flux density (B) increases

(c) the passing current intensity (I) in the coil increases

(d) the net magnetic flux (ϕ_m) through the coil increases

90 The opposite figure represents a rectangular metal frame that carries an electric current while being in the plane of the page parallel to a uniform magnetic field, so the direction of the magnetic dipole moment of that coil is ..... .



(a) parallel to page and rightward (b) parallel to page and leftward

(c) perpendicular into the page (d) perpendicular out of the page

91 A coil of cross-sectional area 0.001 m^2 carrying an electric current of 10 A is placed in a magnetic field of density 2 T and inclined by an angle of 60° to the field. So, the torque acting on it equals 1 N.m, so:

(i) The number of turns of the coil equals ..... turns.

(a) 15 (b) 50 (c) 100 (d) 200

(ii) The maximum value of torque acting on the coil is ..... .

(a) 2 N.m (b) 1.5 N.m (c) 1 N.m (d) 0.5 N.m

(iii) The magnetic dipole moment of the coil equals ..... .

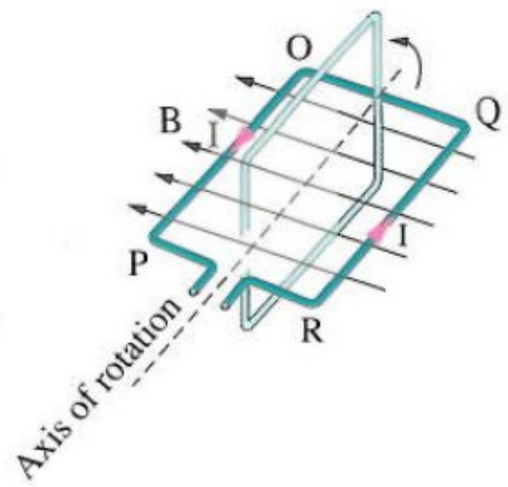
(a) 2 $A.m^2$ (b) 1.5 $A.m^2$ (c) 1 $A.m^2$ (d) 0.5 $A.m^2$

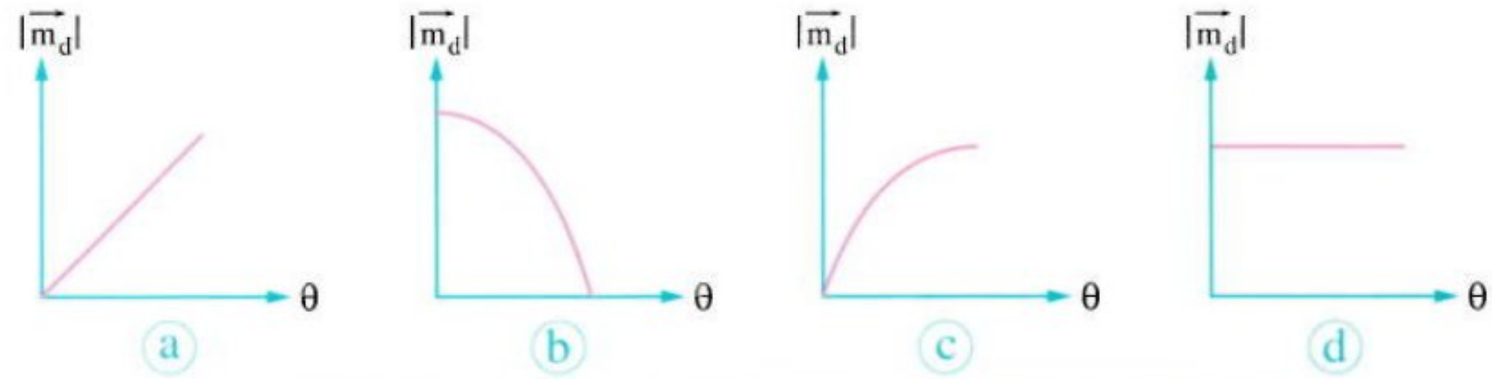
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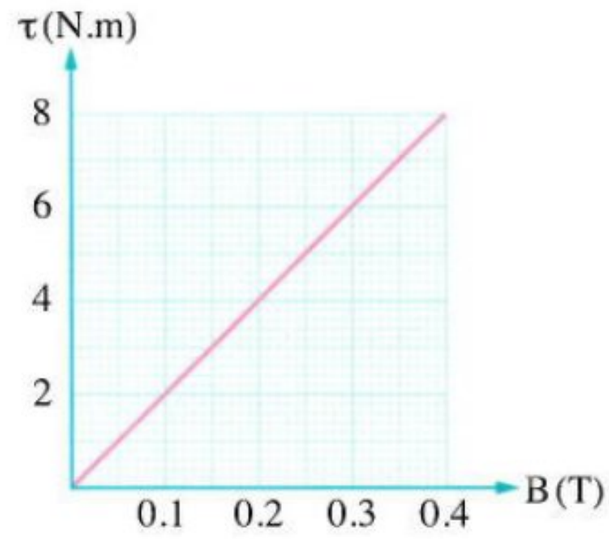
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92 The opposite figure represents a rectangular coil (POQR) that has N turns of wire and carries an electric current of intensity I. The coil is placed in a uniform magnetic flux of density B where the coil's plane is parallel to the magnetic flux lines, so which of the following graphs represents the variation of the magnitude of the magnetic dipole moment $|\overrightarrow{m_d}|$ of the coil with respect to the angle of rotation (θ) when the coil rotates through an angle of 90° from this position?



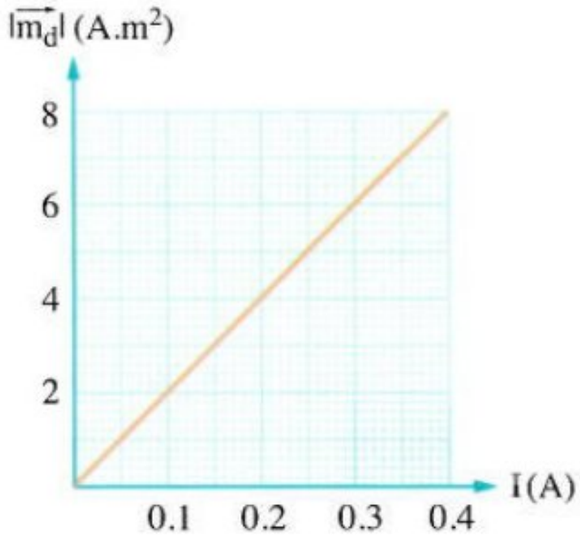


93 The opposite graph represents the relation between the torque (τ) acting on a current-carrying rectangular coil and the magnetic field (B) that is positioned parallel to the plane of the coil and has a changeable uniform strength, so the magnitude of the magnetic dipole moment of the coil equals ..... .



(a) 10 A.m^2 (b) 15 A.m^2
(c) 20 A.m^2 (d) 40 A.m^2

94 A coil of 50 turns carries an electric current and rotates in a uniform magnetic field of flux density B. The opposite graph represents the variation of the magnetic dipole moment $|\overrightarrow{m_d}|$ of the coil with the electric current intensity (I) that passes through it, if a maximum magnetic flux of 0.2 Wb passes through the coil during its rotation, then the magnetic flux density (B) in which the coil is placed is equal to ..... .



(a) 0.5 T (b) 1 T (c) 1.5 T (d) 2 T

95 * A circular coil has N turns of wire and has a radius of 10 cm, if an electric current I passes through it, a magnetic flux of density 2×10^{-4} T is formed at its center, so the magnitude of the magnetic dipole moment of the coil is ..... .

(a) 1 A.m^2 (b) 2 A.m^2 (c) 3 A.m^2 (d) 4 A.m^2

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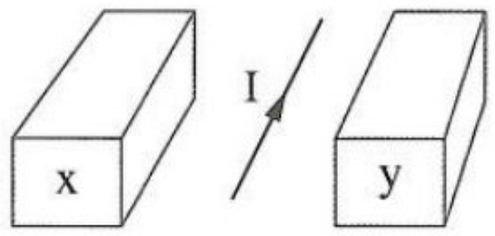
96 * A circular coil of 3.14 cm^2 face area carries a current of intensity I, so the magnetic flux density at its center is 2×10^{-5} T, so its magnetic dipole moment equals ..... .
(Take: $\pi = 3.14$)

(a) 10^{-2} A.m^2 (b) 10^{-4} A.m^2 (c) 10^{-6} A.m^2 (d) 10^{-8} A.m^2

Second Essay questions

The force affecting on a wire

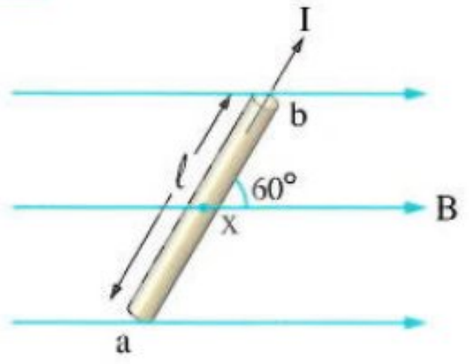
1 A movable straight wire carries an electric current I while being placed horizontally between the poles of magnets x, y as shown in the figure. It is noticed that the wire moves vertically downwards, **determine with explanation** the type of the magnetic pole x.



2 In the opposite figure, a straight wire ab is affected by a magnetic force of 0.1 N, **calculate** the value of the magnetic force that affects the wire if it is rotated about the axis x in the plane of the page by an angle of 30°:

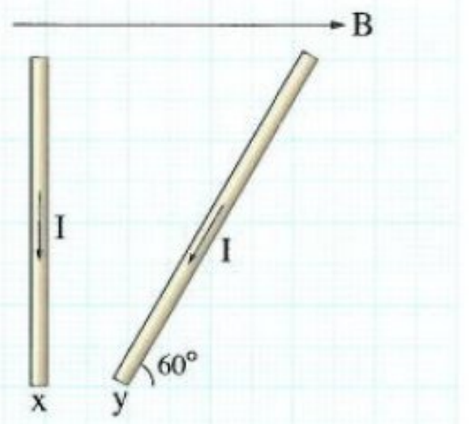
(a) In the clockwise direction.

(b) In the anti-clockwise direction.



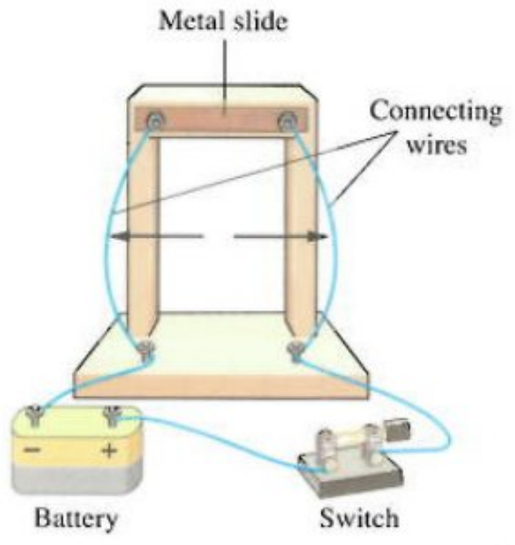
3 A straight wire of resistance 1 Ω for every meter is connected to DC electric source of emf 2 V and its internal resistance is negligible. The wire is placed perpendicular to a magnetic field (B) of flux density 0.1 T, **calculate** the value of the magnetic force that acts on the wire due to the magnetic field (B).

4 In the opposite figure, two straight wires x, y each carries an electric current I. They are placed in a magnetic field of flux density (B) parallel to the common plane of the two wires. **Find** the ratio between the magnetic forces which are acting on them $\left(\frac{F_x}{F_y}\right)$.



The mutual force between wires

5 In the shown experiment, the connecting wires bend outward when an electric current passes through them, **explain this observation**.

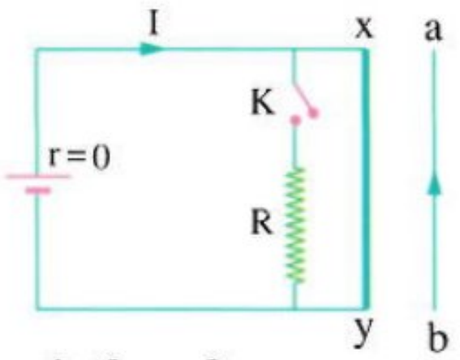


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6 In the opposite figure:

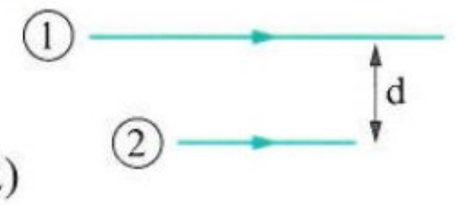
If the resistance of wire xy is R and the passing electric current through the circuit is I:



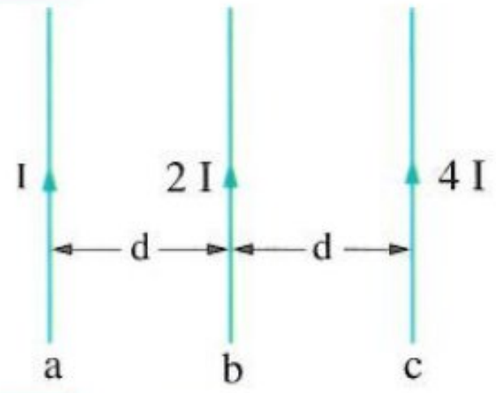
(a) **What** is the type of the mutual electromagnetic force between the two wires (xy and ab) in the case of opening switch K?

(b) When switch K is closed, **what will happen to** this electromagnetic force?

7 In the opposite figure, two parallel straight wires (1), (2) of lengths $\ell, \frac{\ell}{2}$ respectively where each of them carries electric current I. If the magnetic force acting on unit length of the wire (2) is 0.2 N/m, **mention with explanation what happens to** the value of that force when wire (1) is replaced with another of length 2ℓ while keeping both the currents and the same distance between them constant.

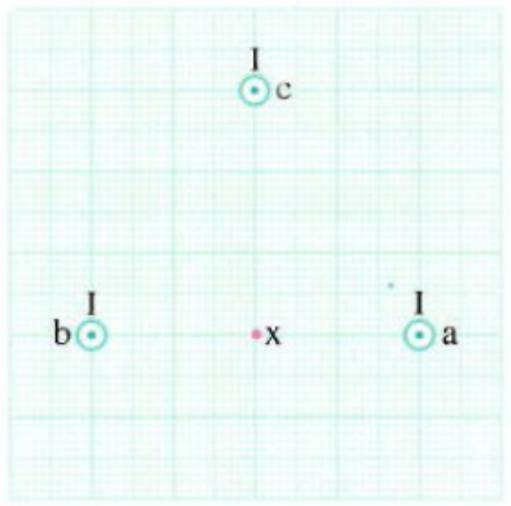


8 Three straight parallel wires a, b, c are in the same plane, each of them carries an electric current as in the opposite figure, **prove that** the wire which is affected by the greatest magnetic force for every unit length is wire c.



9 In the opposite figure, three straight vertical wires a, b, c each of them carries electric current of intensity I.

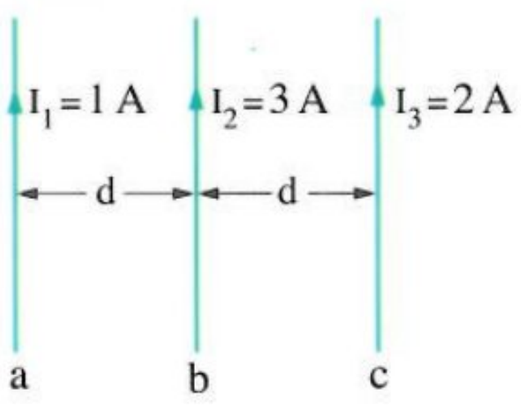
Show what happens to the value of the magnetic force per unit length of wire c if it is transferred to position x.



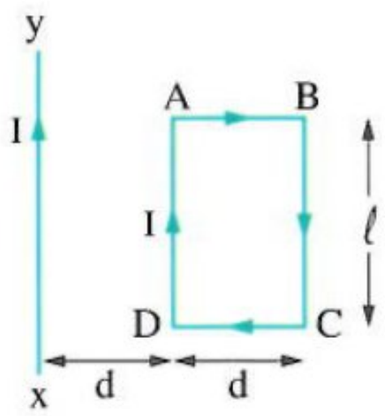
10 Three straight parallel wires a, b, c each of them carries an electric current as in the opposite figure, **what happens to the magnitude and direction of the magnetic force for every unit length of wire a at the following cases:**

(a) reversing the current in wire b?

(b) tripling the current of wire c?

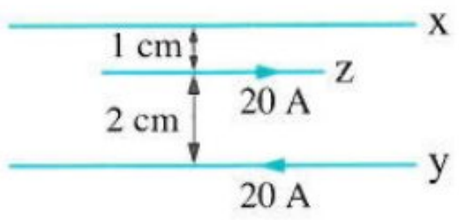


11 In the opposite figure, a rectangular metal frame (ABCD) of length ℓ and width d, carries an electric current of intensity I. A long wire xy in the same plane of the frame and at a distance d from it carries an electric current of intensity I. **Prove that** the total magnetic force acting on the frame can be calculated from the relation; $F = \frac{\mu I^2 \ell}{4\pi d}$



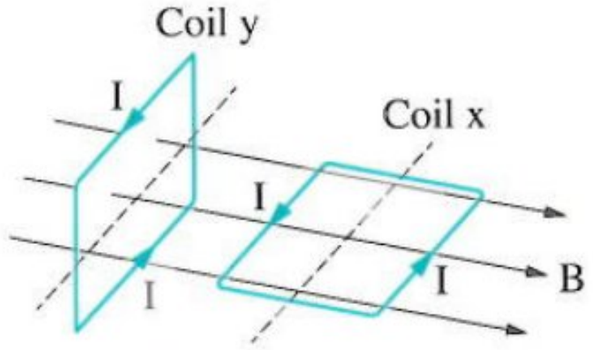
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12 Two long horizontal wires x, y extend in one vertical plane and carry electric currents I, 20 A respectively. A free movable wire z of length 1 m and weight 0.01 N that carries a current of intensity 20 A is placed in the plane of the two wires as in the opposite figure. **Determine** the intensity and the direction of the current passing in wire x that makes wire z stay balanced.

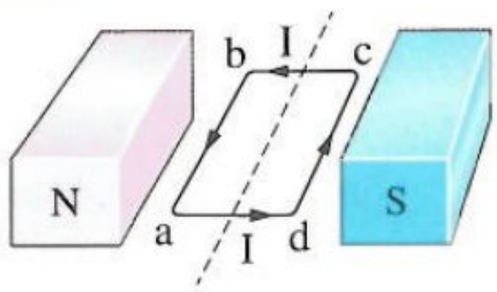


The magnetic torque

13 Two identical coils x, y are placed in a uniform magnetic field (B) as in the opposite figure. When electric current of intensity (I) passes through each of them, coil x rotates but coil y does not. **Explain that.**



14 A rectangular coil abcd that can rotate about its longitudinal axis which is parallel to ab carries a DC electric current and placed parallel to a uniform magnetic field as in the opposite figure:

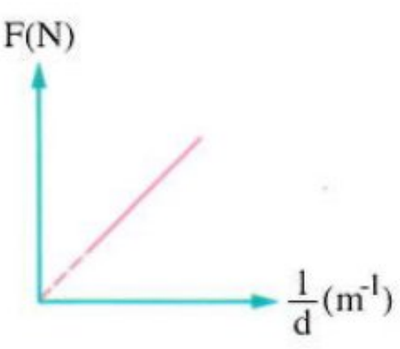
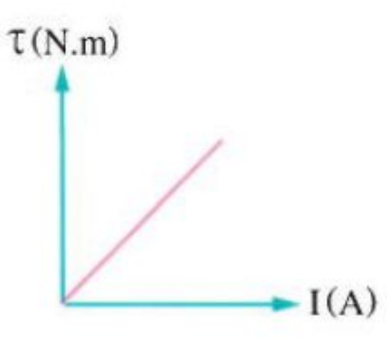
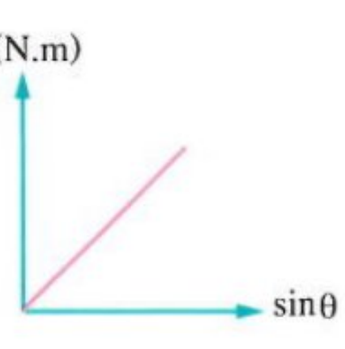


(a) **Determine** the direction of rotation of the coil relative to the clockwise direction.

(b) **What** are the coil sides that are affected by a magnetic force after its rotation for a quarter of a cycle?

15 A ring is placed in a uniform magnetic field of density B, so the maximum magnetic flux that can pass through it is 1 Wb. **Calculate** the maximum torque by which this field can affect the ring when a current of intensity 1 A passes through it.

16 **Write the mathematical relation and the slope for each of the following graphs:**

(a) 	**(b)** 	**(c)** 

Where: (F) is the mutual magnetic force between two parallel wires, (I) is the current, (τ) is the torque on the coil, (θ) is the angle between the normal to the coil and the magnetic field and (d) the distance between the two wires.

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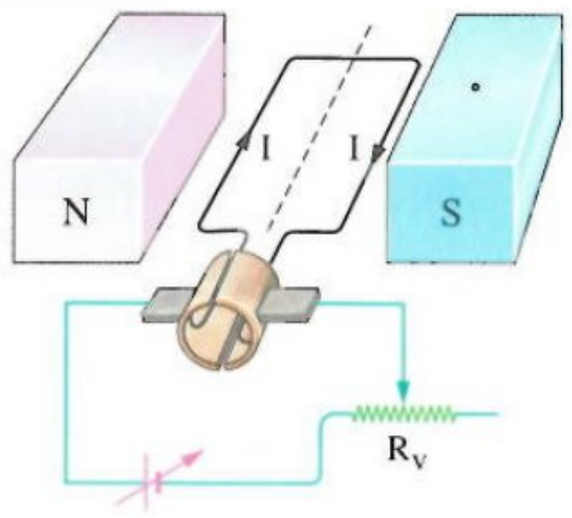
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17 A circular coil of number of turns N where each turn is of area 0.01 m^2 and resistance 1 Ω. The coil is connected to an electric source of emf of 2 V and negligible internal resistance. The coil is placed parallel to a magnetic flux of density 1T, **determine** the torque that acts on it.

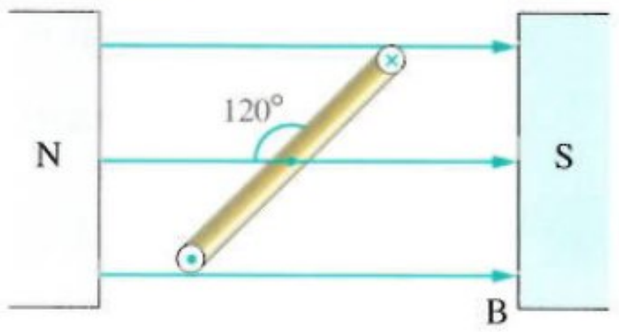
18 A straight wire of length ℓ is coiled in the form of a circular coil of radius r, number of turns N and placed parallel to the direction of a magnetic flux density B. If a current of intensity I is passed in the coil, **prove that** the maximum value of the acting magnetic torque that acts on the coil can be calculated from the relation; $\tau = \frac{1}{2}$ BIrℓ

19 Two straight identical wires, each of them is reshaped in the form of a circular coil where the number of turns of the first is three times that of the second. **Determine** the ratio between the values of the magnetic dipole moment for the two coils if a current of intensity I is passing through each of them.

20 A rectangular coil connected in a closed circuit is placed between the poles of a magnet as shown in the opposite figure. **How** can each of the magnetic dipole moment and the maximum value of torque which are acting on it be increased without changing the specifications of the coil?



21 The opposite figure represents a front view of a rectangular coil of magnetic dipole moment 10 A.m^2 that is placed in a uniform magnetic field of density 0.1 T. **Calculate:**



(a) The torque that acts on the coil in the shown position in the figure.

(b) The torque after rotating the coil 60° in the clockwise direction from the shown position in the figure.

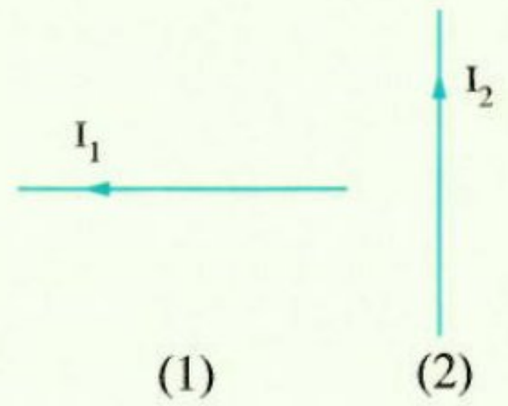
22 A straight wire of length (ℓ) is coiled in the form of a square coil of number of turns N. If a current of intensity (I) is set up through the coil, **prove that** the magnetic dipole moment of the coil can be determined from the relation; $|\overrightarrow{m_d}| = \frac{I\ell^2}{16\,N}$

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Questions of the Previous Years Exams ? on Lesson 3

The magnetic force affecting a wire

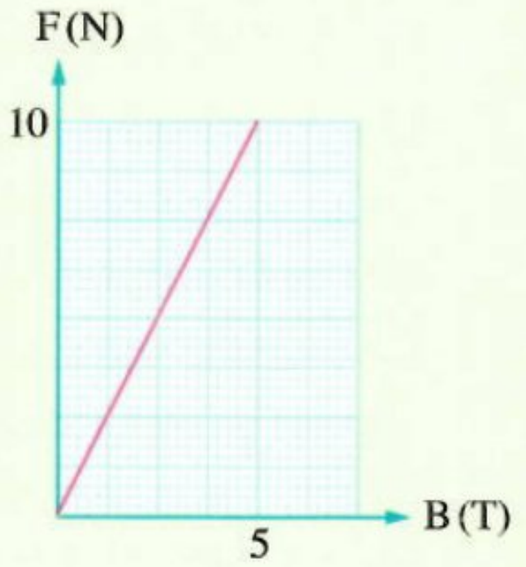
1 In the opposite figure, two straight wires (1) and (2) are in one plane, perpendicular to each other, while two electric currents I_1 and I_2 are set up through them respectively, so the direction of the magnetic force that affects wire (1) due to the effect of the magnetic field arising from the current in wire (2) is ..... .



(a) up the page
(b) down the page
(c) perpendicular into the page
(d) perpendicular out of the page

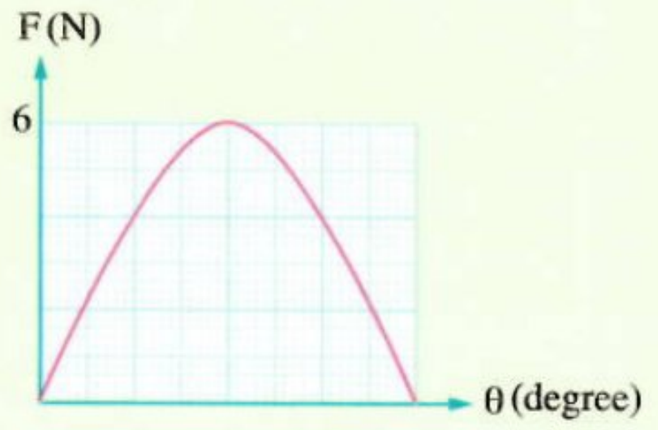
(Exp. May 21)

2 The graph represents the relation between the magnetic force (F) acting on a current-carrying wire when it is placed perpendicular to a changing magnetic field and the magnetic flux density (B) of the field, so the magnetic force acting on the wire when the magnetic flux density equals 3 T will be ..... newtons. *(Exp. June 21)*



(a) 6 (b) 4
(c) $\frac{1}{2}$ (d) 2

3 The opposite graph represents the relation between the magnetic force (F) acting on a current-carrying straight wire that is placed in a magnetic field of density (B) and the angle between the direction of the magnetic flux and the wire (θ). So, the angle (θ) at which the magnetic force (F) that acts on the wire equals half its maximum value is ..... . *(1st session 21)*

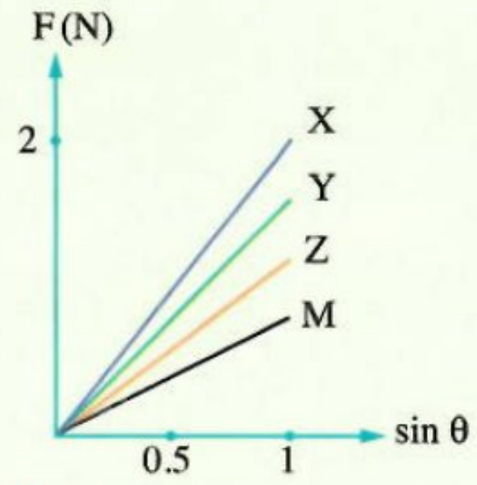


(a) 30° (b) 45° (c) 60° (d) 120°

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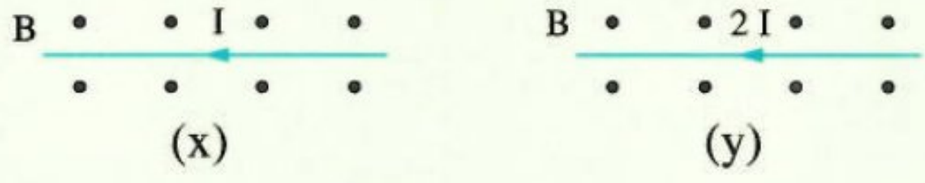
4 (X), (Y), (Z) and (M) are four straight wires that are different in length, each carries electric current of intensity I and all are placed in the same magnetic field of flux density B. The graph represents the relation between the magnetic force (F) that affects each wire and sine of the angle between each wire and the direction of the magnetic flux (sin θ). So, the longest wire is ..... .



(2nd session 21)

(a) wire (X) (b) wire (Y) (c) wire (Z) (d) wire (M)

5 Two straight wires (X) and (Y) of the same length are carrying electric currents in the directions shown in the figure. They are placed perpendicular to a uniform magnetic flux of density (B) perpendicular to and out of the page.



(x) (y)

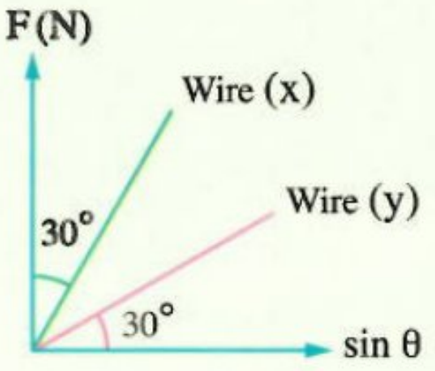
So, the relation between the magnetic force [F_x] that is acting on wire (x) and the magnetic force [F_y] that is acting on wire (y) is ..... .

(1st session 22)

(a) $F_y > F_x$ and both are directed downwards (b) $F_y > F_x$ and both are directed upwards

(c) $F_x > F_y$ and both are directed upwards (d) $F_x > F_y$ and both are directed downwards

6 The graph represents the relation between the magnetic force (F) acting on two straight wires (x, y) and sine the angle between each wire and the direction of the magnetic field of flux density (B) in which the wires are placed (sin θ). If the ratio between:



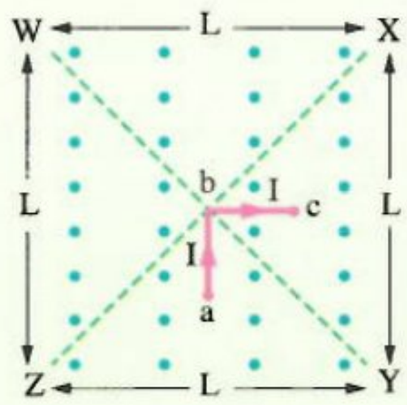
$$\frac{\text{electric current intensity passing in the wire (x)}}{\text{electric current intensity passing in the wire (y)}} = \frac{3}{4}$$

So, the ratio between : $\frac{\text{the length of wire (x)}}{\text{the length of wire (y)}}$ = ...

(2nd session 22)

(a) $\frac{4}{3}$ (b) $\frac{4}{9}$ (c) $\frac{4}{1}$ (d) $\frac{8}{3}$

7 A straight metal wire that carries an electric current of (I) intensity, is bent into two equal and perpendicular parts {ab, bc}, then placed parallel to the page inside a uniform magnetic field which is perpendicular out of the page as in the figure, toward which point (Z, Y, X, W) will point b of the wire move?

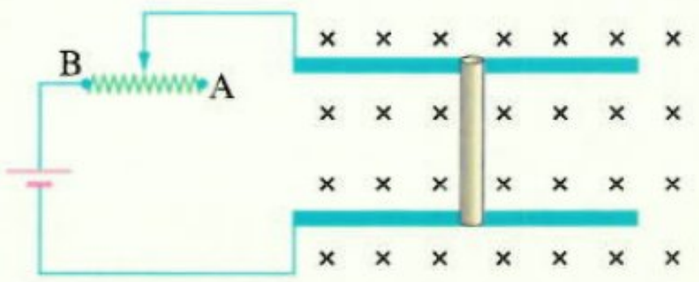


(a) Point Y. (b) Point X.

(c) Point W. (d) Point Z. *(2nd session 22)*

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8 A cylindrical metallic rod, slides on two copper sheets fixed in the same plane of the page, the two sheets are connected to a battery and a rheostat where the rod and sheets are placed in a uniform magnetic field of flux lines perpendicular to the plane of the page as in figure, so a magnetic force F acts on the rod.



Which of the following represents. What happens to the value and the direction of the force F when the slider of the rheostat moves towards point B?

(Exp. 23)

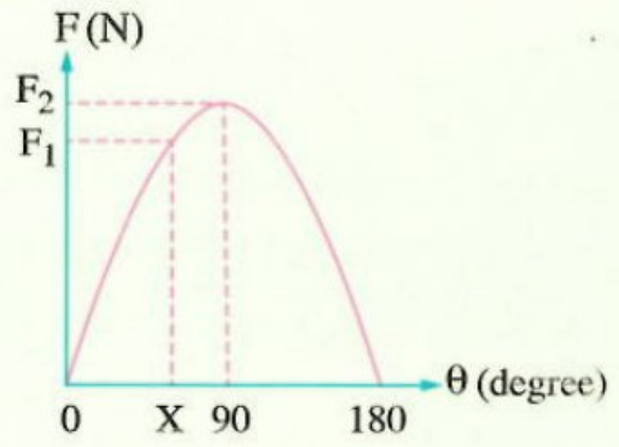
(a) Decreases and remains in the same direction.

(b) Increases and remains in the same direction.

(c) Decreases and reverses its direction.

(d) Increases and reverses its direction.

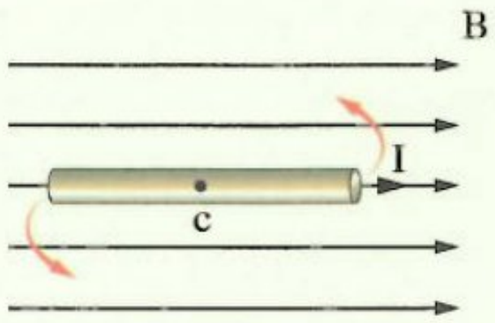
9 The graph represents the relation between the magnetic force (F) acting on a straight wire of length (ℓ) carrying an electric current of intensity I placed in a uniform magnetic field of flux density (B) and the change of angle (θ) which is the angle between the wire and the field. If $\frac{F_2}{F_1} = \frac{2\sqrt{3}}{3}$, so the value of angle (X) = ...



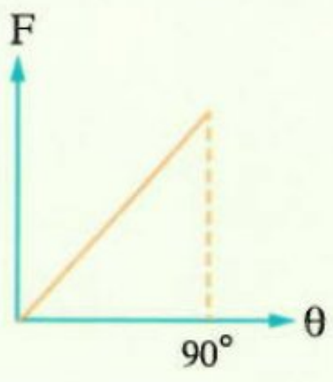
(2nd session 23)

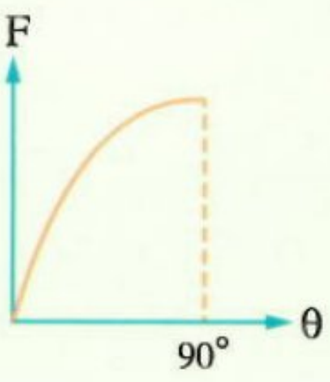
(a) 45° (b) 75° (c) 60° (d) 80°

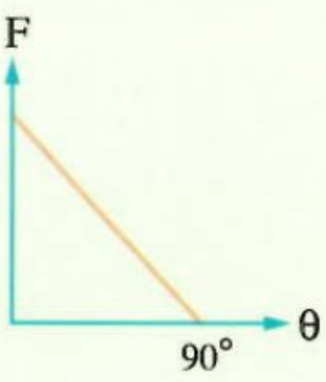
10 In the opposite figure, a straight wire carries an electric current (I) and is placed parallel to a uniform magnetic field of flux density B. If the wire rotates a quarter cycle around an axis perpendicular to the plane of the page at point c, the graph that represents the relation between the magnetic force (F) acting on the wire and the angle of rotation (θ) is ..... .

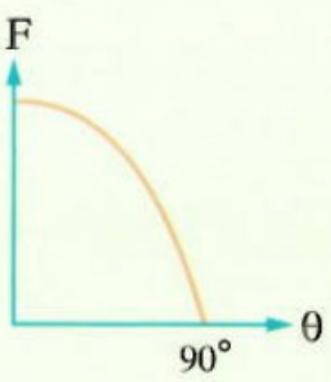


(Practice 25)

(a)


(b)


(c)


(d)


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11 A wire of weight F is horizontally suspended inside a uniform magnetic field that is acting perpendicular to it, of flux density B as shown in the opposite figure. When an electric current of intensity I passes through the wire, it is affected by a magnetic force of magnitude 2 F. The net force acting on the wire equals ..... .



(Practice 25)

(a) F (b) 2 F (c) $\sqrt{5}$ F (d) 3F

12 A solenoid contains 250 turns/m and carries a current of intensity 5A. If a wire of length 0.35 m that carries a current of 10 A is placed coinciding with the axis of the solenoid, the magnetic force acting on the wire equals ..... .

(Practice 25)

(a) zero (b) 5.5×10^{-3} N (c) 7.5×10^{-3} N (d) 0.01 N

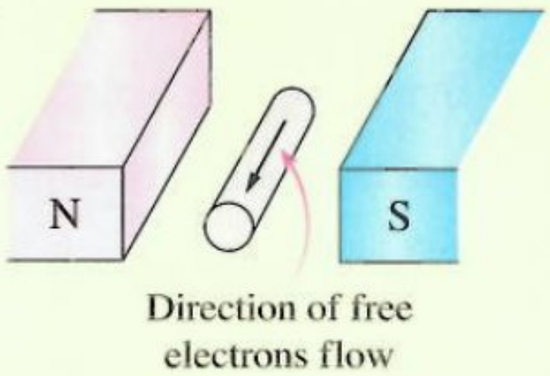
13 An electron moves perpendicularly out of the plane of the page into a uniform magnetic field directed southward in the plane of the page. Then the direction of the magnetic force acting on the moving electron is ..... .

(Practice 25)

(a) parallel to the page eastwards (b) parallel to the page westwards
(c) parallel to the page northwards (d) perpendicular into the plane of the page

14 In the figure, a horizontal straight wire is placed perpendicular to the direction of the magnetic flux of a magnet.

When free electrons flow through the wire, the wire experiences a magnetic force because ..... .



(a) of a difference in magnetic flux density above and below the wire
(b) free electrons within it are attracted to the north pole of the magnet
(c) free electrons within it are attracted to the south pole of the magnet
(d) of a difference in magnetic flux density at the right and left of the wire *(1st session 25)*

15 The figures below show a straight wire carrying an electric current of intensity (I) at different positions in the plane of the page, in a uniform magnetic field perpendicular to the page.



Position (1) **Position (2)** **Position (3)**

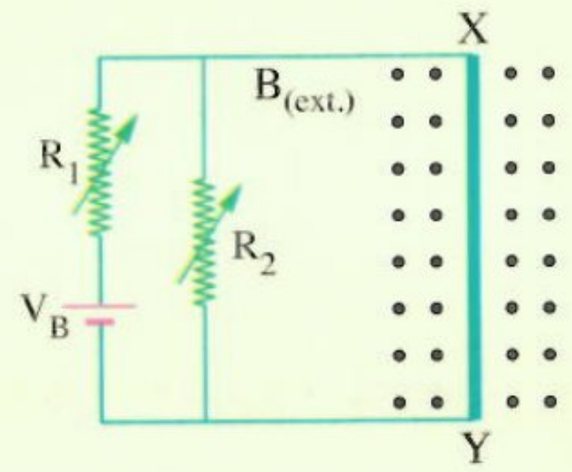
So, the magnetic force acting on the wire is ..... .

(2nd session 25)

(a) equal in the three positions (b) maximum in position (1)
(c) maximum in position (2) (d) maximum in position (3)

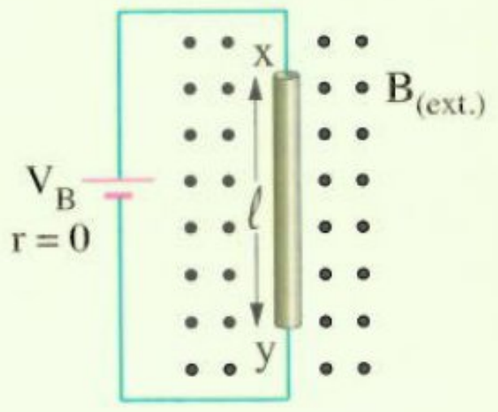
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16 In the shown electrical circuit, a metal wire XY of resistance (R) is connected to a closed circuit containing a battery of electromotive force V_B and negligible internal resistance, and two variable resistors (R_1) and (R_2). The wire is placed perpendicularly in a uniform external magnetic field (B). Which of the following actions individually for each of the values of the two variable resistors R_1 and R_2, respectively causes a decrease in the magnetic force acting on wire XY? *(Practice 26)*



(a) increases, increases.
(b) increases, decreases.
(c) decreases, increases.
(d) decreases, decreases.

17 In the shown figure, a wire xy of length (ℓ) and resistance (R) having a uniform cross-section is connected to a battery of electromotive force (V_B) and of negligible internal resistance and placed in a uniform magnetic field perpendicular out of the plane of the page, with a flux density B producing magnetic force acting on the wire equals F. If the wire has been drawn uniformly to become of length 2 ℓ and placed entirely in the same field and connected to the same battery, the magnitude of magnetic force acting on it ..... . *(Practice 26)*

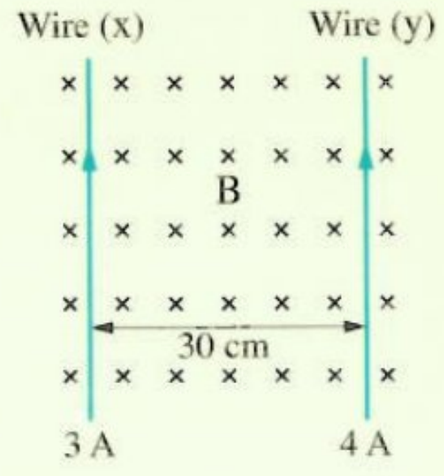


(a) is halved
(b) is quartered
(c) is doubled
(d) remains unchanged

The mutual force between two wires

18 The figure shows two parallel wires (x) and (y), separated by normal distance 30 cm, carrying electric currents of intensities 3 A and 4 A respectively and being affected by an external magnetic field of flux density B that is directed normal into the page as shown. If the resultant magnetic force per unit length of the wire (x) is 2×10^{-5} N/m, the value of B equals ..... . *(Exp. June 21)*

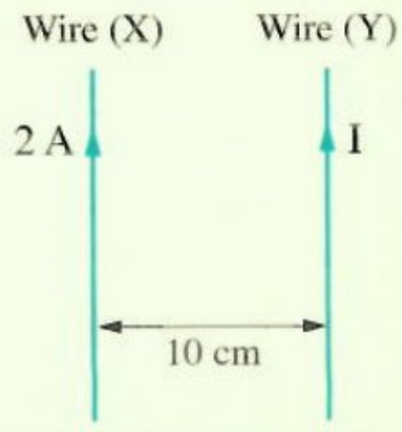
(Take: $\mu = 4\pi \times 10^{-7}$ T.m/A)



(a) 6.67×10^{-6} T
(b) 9.3×10^{-6} T
(c) 4×10^{-6} T
(d) 2.67×10^{-6} T

19 The figure represents two parallel straight wires (X) and (Y). If the force acting on a unit length of each wire is 4×10^{-5} N/m, the passing electric current intensity (I) in wire (Y) equals ..... .

(Take: $\mu = 4\pi \times 10^{-7}$ Wb/A.m) *(1st session 21)*



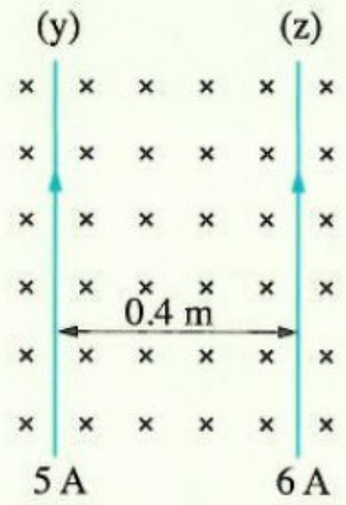
(a) 0.1 A
(b) 1 A
(c) 10 A
(d) 100 A

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20 The figure shows two parallel straight wires (z), (y) separated by a normal distance of 0.4 m, carrying electric currents of intensities 6 A and 5 A respectively and being affected by external magnetic field of flux density 2.5×10^{-5} T perpendicular into the page as shown in the figure, so the net magnetic force that affects unit length of the wire (z) is approximately equal to ..... *(2nd session 21)*

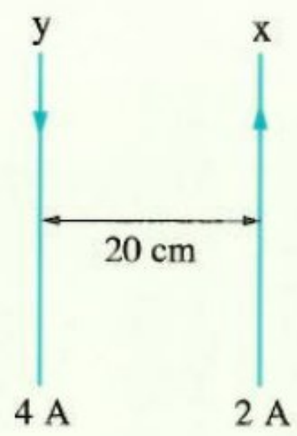
(Knowing that: $\mu = 4\pi \times 10^{-7}$ T.m/A)



(a) 1.5×10^{-5} N/m
(b) 1.5×10^{-4} N/m
(c) 1.7×10^{-4} N/m
(d) 4×10^{-5} N/m

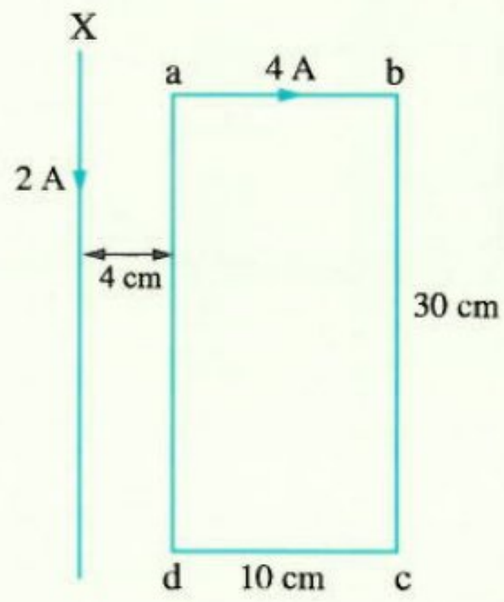
21 The figure shows two parallel straight wires (x), (y), each of length 1.6 m and the normal distance between them is 20 cm, carrying electric currents of intensities 2 A and 4 A respectively. The mutual magnetic force between (x, y) is ..... .

($\mu = 4\pi \times 10^{-7}$ T.m/A) *(1st session 22)*



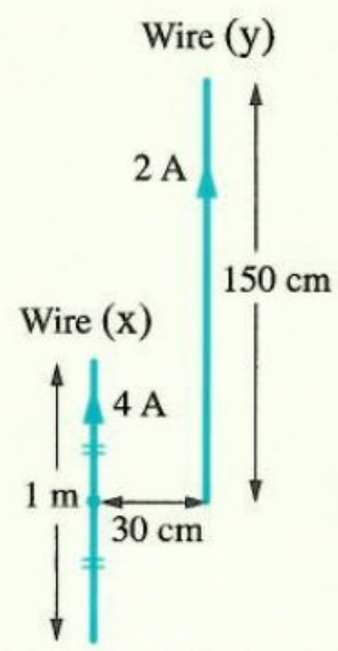
(a) 1.28×10^{-4} N
(b) 1.28×10^{-6} N
(c) 1.28×10^{-7} N
(d) 1.28×10^{-5} N

22 The figure shows a conducting rectangle (abcd) that carries 4 A electric current, another wire (X) which carries 2 A electric current is placed 4 cm away from it, so the magnitude and the direction of the magnetic force affecting wire X are ..... .



(a) 1.54×10^{-5} N, to the left
(b) 1.54×10^{-5} N, to the right
(c) 8.57×10^{-6} N, to the right
(d) 8.57×10^{-6} N, to the left *(1st session 22)*

23 Two current-carrying straight wires x, y are parallel as shown in the figure, the mutual magnetic force between the two wires equals ..... . ($\mu = 4\pi \times 10^{-7}$ tesla.m/A) *(2nd session 22)*

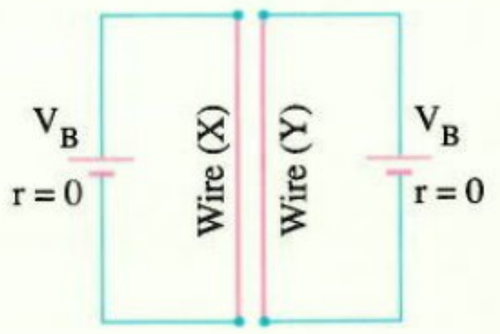


(a) 2.67×10^{-6} N
(b) 8×10^{-6} N
(c) 5×10^{-6} N
(d) 5.33×10^{-6} N

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24 Two long parallel wires X and Y, each of them is connected to a battery of negligible internal resistance, the mutual force between them is (F).
When wire X is replaced by another one of the same length and radius but the resistivity of its material is $\frac{1}{4}$ of that of wire X, so the mutual force between the two wires becomes ..... .

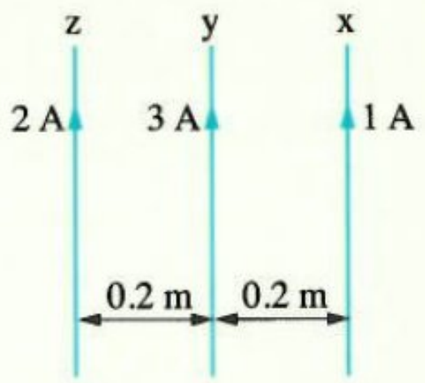
(Exp. 23)



(a) 2 F (b) F (c) 4 F (d) $\frac{F}{4}$

25 From the data in the figure, which of the following choices represents the correct arrangement of magnetic forces per unit length acting on each wire?

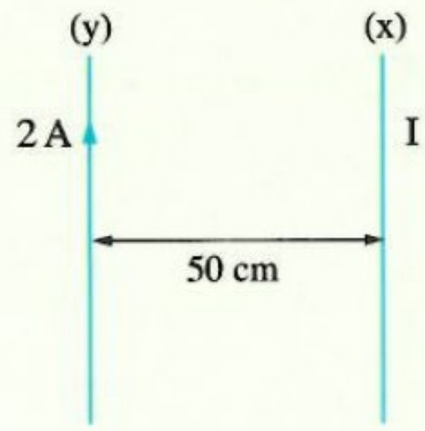
(Exp. 23)



(a) $F_y < F_x < F_z$ (b) $F_z < F_y < F_x$
(c) $F_x < F_y < F_z$ (d) $F_y < F_z < F_x$

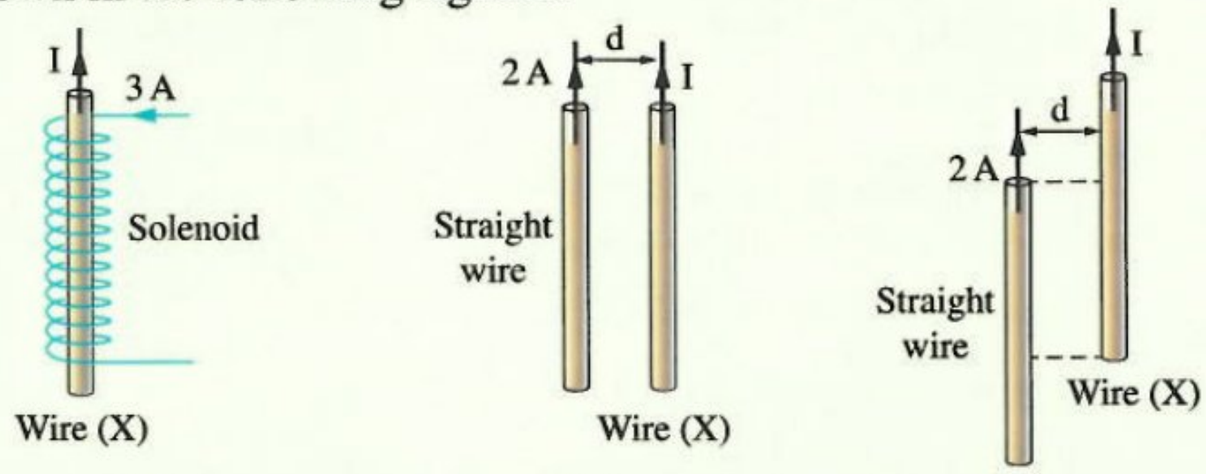
26 In the shown figure:
If wire (x) is affected by a magnetic force per unit length of 2×10^{-6} N/m rightwards due to the effect of the magnetic flux resulted due to the passing current through wire (y), then the intensity and the direction of current (I) are ..... .
($\mu = 4\pi \times 10^{-7}$ T.m/A)

(1st session 23)



(a) 2.5 A, upward (b) 2.5 A, downward
(c) 25 A, downward (d) 25 A, upward

27 A wire (X) that carries a current of intensity (I) is placed in different magnetic fields as shown in the following figures.



Which of the following choices represents the correct arrangement of the values of the magnetic force acting on the wire (X) according to each figure?

(1st session 23)

(a) $F_2 > F_3 > F_1 = F_4$ (b) $F_2 = F_3 > F_1 = F_4$
(c) $F_1 > F_2 > F_3 > F_4$ (d) $F_1 > F_2 = F_3 = F_4$

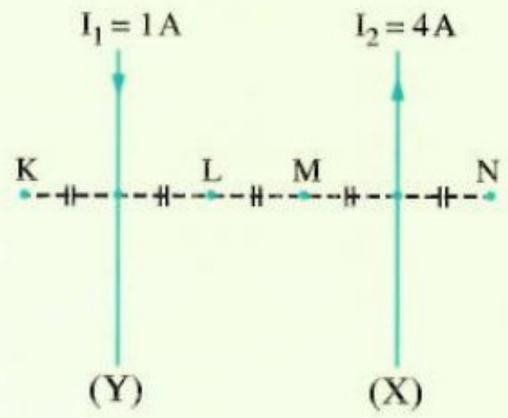
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28 Two parallel straight wires A and B, carry electric currents of intensities 2 A and I respectively in opposite directions. If the normal separating distance between the two wires is 5 cm and the common length between them is 10 cm. The two wires are affected by a mutual magnetic force of 2.4×10^{-6} N, then the value of (I) should be ..... and the type of the mutual force is ..... .
(Take: $\mu = 4\pi \times 10^{-7}$ Wb/A.m) *(2nd session 23)*

(a) 3 A, repulsion
(b) 3 A, attraction
(c) 6 A, repulsion
(d) 6 A, attraction

29 From the shown figure, at which point a straight current carrying wire can be placed in the same plane of the page, parallel to the two wires (X) and (Y) and is not affected by a magnetic force? *(1st session 24)*

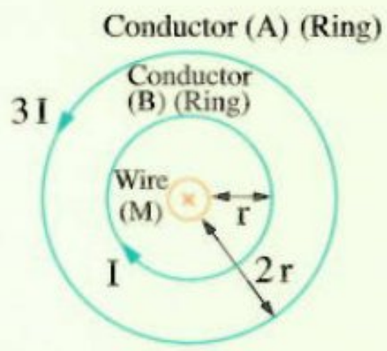


(a) K
(b) L
(c) M
(d) N

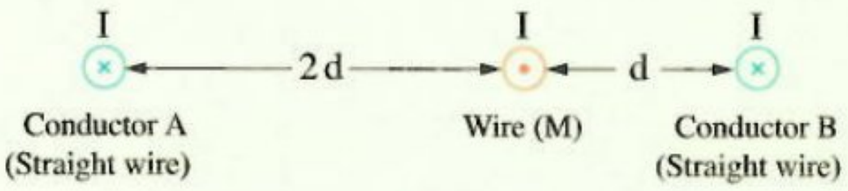
30 A current carrying wire (M) is placed perpendicular to the plane of the page surrounded by two different current carrying conductors (A and B), so in which figure the wire (M) will not be affected by a magnetic force due to the magnetic field produced from the surrounding conductors? *(1st session 24)*



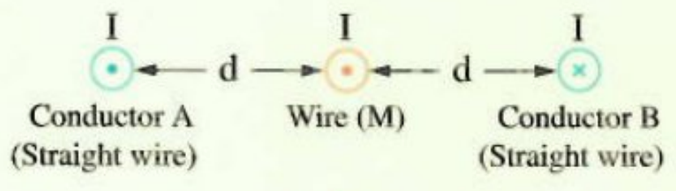
(a)



(b)

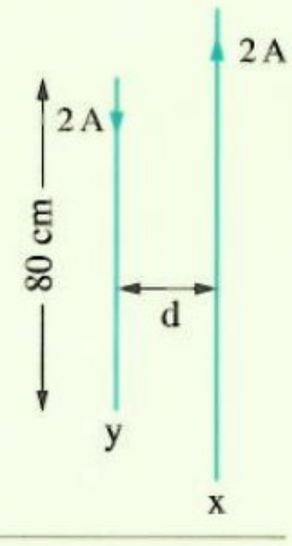


(c)



(d)

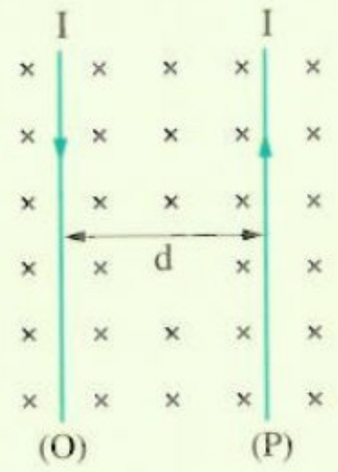
31 The figure illustrates two parallel straight wires (x), (y), wire (x) is very long and wire (y) has a length of 80 cm and both are carrying electric currents as shown. The mutual force between them is 2×10^{-5} N so, the normal distance (d) between the two wires equals ..... .
($\mu = 4\pi \times 10^{-7}$ T.m/A) *(2nd session 24)*



(a) 3.2 cm
(b) 0.32 cm
(c) 0.032 cm
(d) 0.0032 cm

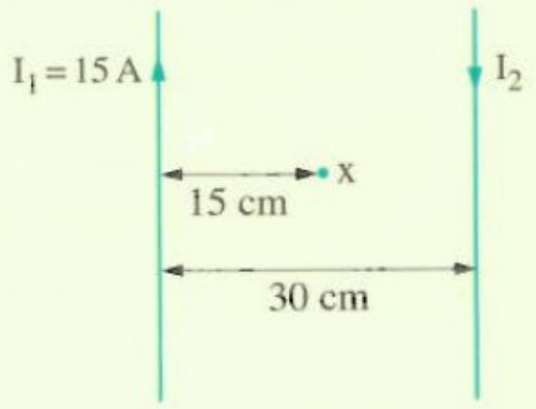
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32 Two long parallel wires (O) and (P) are in the same plane of the page, separated by a distance (d) and two electric currents of the same intensity (I) pass through them in opposite directions and affected by uniform magnetic field of flux density $\frac{\mu I}{\pi d}$ as shown in the figure, if the wire (P) is free to move but wire (O) is fixed, in which direction will the magnetic force act on wire (P)? *(2nd session 24)*



(a) Not affected by a magnetic force.
(b) Towards the left of the page.
(c) Towards the right of the page.
(d) Perpendicular to the plane of the page.

33 In the opposite figure, two long parallel straight wires are separated by a distance of 30 cm. The first wire carries a current of 15 A. If the resultant magnetic flux density at point x, which is mid-point between the two wires, equals 6×10^{-5} T, then the mutual magnetic force per unit length between the wires acting on either of them equals ..... . *(Practice 25)*



(a) 2×10^{-4} N
(b) 3×10^{-4} N
(c) 4×10^{-4} N
(d) 5×10^{-4} N

34 Figure (1) shows two straight parallel wires (X) and (Y) carrying electric currents 2I and I respectively. The mutual magnetic force between the two wires is F. When both wires are placed in an external magnetic field of flux density B, as shown in figure (2), if the net magnetic force acting on wire (X) vanishes. The net magnetic force acting on wire (Y) is ..... . *(Practice 25)*

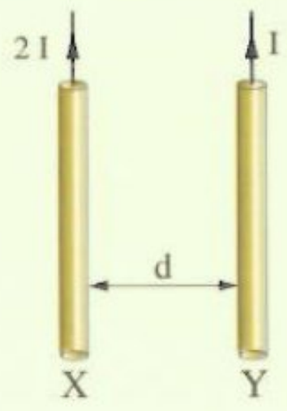


Figure (1)

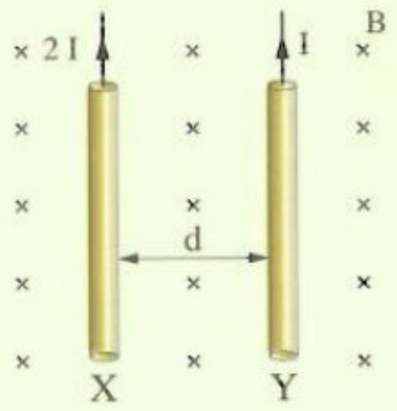
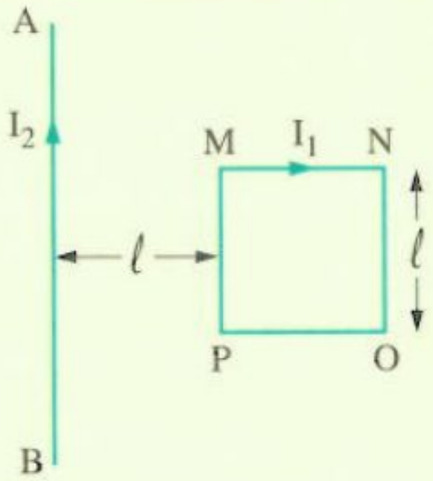


Figure (2)

(a) 0.5 F
(b) 1 F
(c) 1.5 F
(d) 2 F

35 A freely movable square coil MNOP consists of one turn carries an electric current of intensity I_1, a long straight fixed wire (AB) is placed near to the coil and in its same plane and also carries a current of intensity I_2, as shown in the opposite figure. If the net magnetic force acting on the wire equals F, considering the following statements: *(Practice 25)*



(I) The magnitude of the mutual magnetic force between the wire AB and the side MP equals 2 F.
(II) The magnitude of the mutual magnetic force between the wire AB and the side NO equals F.

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(III) The coil rotates in the anti-clockwise direction.

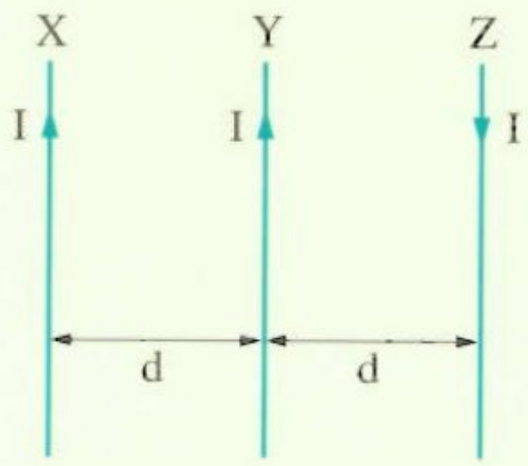
(IV) The coil moves approaching the straight wire.

Which of the previous statements is correct?

(a) (I), (II), (IV) only. (b) (I), (II) only.

(c) (II), (IV) only. (d) (I), (II), (III), (IV).

36 Three identical straight wires, each carries an electric current of intensity (I), are parallel to each other where the perpendicular distance between them is as shown in the figure. Which wire is affected by the least magnetic force?



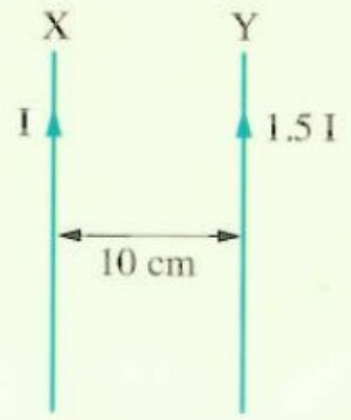
(a) The three wires are affected by equal forces.

(b) Wire (Y).

(c) Wire (Z).

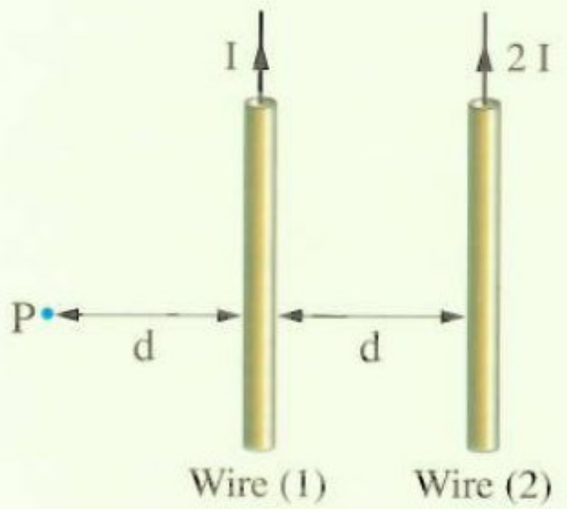
(d) Wire (X). *(1st session 25)*

37 Two straight parallel wires (X and Y) have a perpendicular distance of 10 cm between them. Two electric currents of intensities I and 1.5 I pass through them, respectively. If the magnitude of the mutual magnetic force between the two wires per unit length is 1.2×10^{-5} N/m, the current intensity (I) is equal to ..... .
($\mu = 4\pi \times 10^{-7}$ Wb/A.m)



(a) 2 A (b) 1 A (c) 1.5 A (d) 2.5 A *(2nd session 25)*

38 Two long, parallel, straight wires (1) and (2) carry electric currents as shown in the opposite figure. If the total magnetic flux density at point P equals B, then the magnetic force acting on a unit length of wire (1) equals ..... . *(Practice 26)*



(a) BI (b) 2BI

(c) $\frac{BI}{2}$ (d) $\frac{BI}{4}$

The magnetic torque

39 A circular coil of 30 turns and cross-sectional area 10 cm^2 is placed in a magnetic field of flux density 0.3 T while carrying a current of 2 A. Knowing that the direction of the magnetic dipole moment of the coil makes an angle of 30° with the direction of the magnetic field, so the magnitude of the magnetic torque that affects the coil is ..... .

(a) $9\sqrt{3} \times 10^{-3}$ N.m (b) 18×10^{-3} N.m

(c) $18\sqrt{3} \times 10^{-3}$ N.m (d) 9×10^{-3} N.m *(Exp. May 21)*

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40 An electric current passes through a rectangular coil whose plane is parallel to the direction of a magnetic field of flux density 2 T, if the magnetic dipole moment of coil is 0.3 $A.m^2$, the magnetic torque acting on the coil equals ..... . *(Exp. June 21)*

(a) 0.6 N.m (b) 0.15 N.m (c) 0.06 N.m (d) 0.015 N.m

41 A rectangular coil, that has no. of turns = 2 turns, length =10 cm and width = 2 cm, carries an electric current of 2 A while being placed in a uniform magnetic field of flux density 2 T, so the magnetic torque acting on the coil, when the angle between the coil and the direction of the magnetic flux = 60°, equals ..... . *(1st session 21)*

(a) 16×10^{-3} N.m (b) $8\sqrt{3} \times 10^{-3}$ N.m (c) 8×10^{-3} N.m (d) 16×10^{-4} N.m

42 If the magnetic torque acting on a current-carrying coil while being placed in magnetic field = 0.86 N.m when the angle between the normal to coil plane and the direction of field = 60°, so the magnetic torque acting on the coil when the coil plane becomes parallel to the magnetic field ≈ ..... *(2nd session 21)*

(a) 1 N.m (b) 1.5 N.m (c) 1.86 N.m (d) zero

43 A current-carrying coil is placed inside a magnetic field of flux density 400 mT, where the coil plane makes angle θ with the direction of the magnetic flux. If the ratio of ; $\frac{\text{The value of the magnetic dipole moment}}{\text{The value of the magnetic torque}} = 5\ T^{-1}$, then the value of angle (θ) equals ..... . *(1st session 22)*

(a) 30° (b) 35° (c) 60° (d) 55°

44 A 5 turns rectangular coil of dimensions 20 cm, 40 cm, is placed inside a magnetic field of flux density 0.02 T where the coil plane makes angle 55° with the flux direction. If an electric current of 4 A intensity passes through the coil, then the magnetic torque acting on that coil equals ..... . *(2nd session 22)*

(a) 18.4×10^{-3} N.m (b) 26.2×10^{-3} N.m

(c) 320×10^{-3} N.m (d) 640×10^{-3} N.m

45 A rectangular coil made of an insulated wire, its dimensions are 0.1 m and 0.05 m, consists of 50 turns, is placed in a magnetic flux of 10^{-3} Wb perpendicularly to the coil plane, it can rotate about an axis in the same plane of its surface that is parallel to its length. If a current of 2 A passes through the coil, the value of the magnetic torque acting on the coil is equal to ..... . *(Exp. 23)*

(a) 0.1 N.m (b) zero (c) 5×10^{-4} N.m (d) 2×10^{-3} N.m

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46 A current of intensity (I) passes in a current-carrying coil which is placed in a magnetic field of flux density (B), where the coil plane makes an angle 60° with the direction of the magnetic flux. If the value of magnetic dipole moment in ($A.m^2$) is 4 times as the value of the magnetic torque acting on the coil in (N.m), then the value of the magnetic flux density (B) equals ..... . *(1st session 23)*

(a) 3.46 T (b) 2 T (c) 8 T (d) 0.5 T

47 A rectangular coil carrying a current is placed in a magnetic field where its plane makes an angle 60° with the direction of the magnetic field lines. If the value of the magnetic torque acting on the coil in (N.m) equals the value of the magnetic dipole moment of the coil in ($A.m^2$), the magnetic flux density equals ..... . *(2nd session 23)*

(a) 2 T (b) 1.15 T (c) 0.5 T (d) 0.86 T

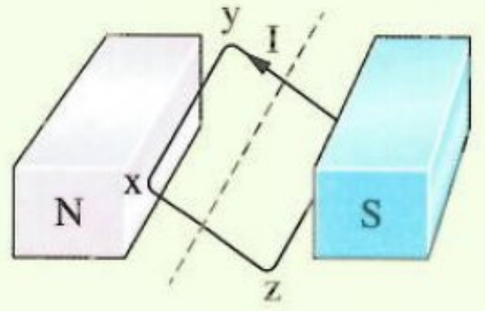
48 A rectangular coil carries an electric current and is inclined at an angle of 30° to the lines of a uniform magnetic field of flux density 0.5 T. If the magnetic dipole moment acting on the coil is 120 $A.m^2$, then the torque acting on the coil is approximately equal to ..... .

(a) 75 N.m (b) 60 N.m (c) 52 N.m (d) 34 N.m *(Practice 25)*

49 A straight wire was shaped into a circular coil consisting of one turn and had an electric current passing through it. It was placed parallel to a uniform magnetic field and affected by a magnetic torque (τ). If the wire is reshaped into a circular coil with 3 turns, placed parallel to the same magnetic field and carrying the same current, the magnetic torque acting on the coil becomes ..... . *(Practice 25)*

(a) 3τ (b) $\frac{\tau}{3}$ (c) 9τ (d) $\frac{\tau}{9}$

50 A rectangular coil carries a direct current. The coil is placed in a uniform magnetic field of flux density 0.5 Wb/m^2 such that its plane makes an angle 60° to the direction of the field, as shown in the figure. The coil is affected by a torque of 14 N.m.



What is the magnitude and direction of the magnetic dipole moment of the coil? *(1st session 25)*

(a) 56 $A.m^2$, perpendicular to the coil plane, and out of the coil.

(b) 56 $A.m^2$, perpendicular to the coil plane, and into the coil.

(c) 32.3 $A.m^2$, perpendicular to the coil plane, and into the coil.

(d) 32.3 $A.m^2$, perpendicular to the coil plane, and out of the coil.

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51 As a direct electric current passes through a rectangular coil, the magnetic dipole moment of the coil equals 56 $A.m^2$. The coil is placed at a certain angle in a uniform magnetic field having flux density 0.5 Wb/m^2. It experiences a torque of 14 N.m. What is the angle between the coil plane and the direction of the magnetic field at that position?

(a) 90° (b) 30° (c) 60° (d) 45° *(2nd session 25)*

52 A straight wire of length (ℓ) is reshaped into a circular loop, then if an electric current of intensity I passes through it, the magnitude of the magnetic dipole moment of the loop ($|\overrightarrow{m_d}|$) is determined by the relation; ..... . *(Practice 26)*

(a) $|\overrightarrow{m_d}| = \frac{I\ell^2}{4\pi^2}$ (b) $|\overrightarrow{m_d}| = \frac{I\ell}{4\pi^2}$ (c) $|\overrightarrow{m_d}| = \frac{I\ell^2}{4\pi}$ (d) $|\overrightarrow{m_d}| = \frac{I\ell}{4\pi}$

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